

UCLA Algebra Qualifying Exam Problems

2000–2026

Organized by Topic and Difficulty

Compiled from UCLA Mathematics Department Archives

September 2026

Contents

Preface	7
1 Group Theory	9
1.1 Standard Problems (★)	9
1.2 Moderate Problems (★★)	10
1.3 Challenging Problems (★★★)	21
2 Ring Theory	23
2.1 Standard Problems (★)	23
2.2 Moderate Problems (★★)	24
2.3 Challenging Problems (★★★)	33
3 Module Theory	35
3.1 Standard Problems (★)	35
3.2 Moderate Problems (★★)	35
3.3 Challenging Problems (★★★)	39
4 Field Theory & Galois Theory	41
4.1 Standard Problems (★)	41
4.2 Moderate Problems (★★)	42
4.3 Challenging Problems (★★★)	52
5 Linear Algebra	57
5.1 Standard Problems (★)	57
5.2 Moderate Problems (★★)	57
5.3 Challenging Problems (★★★)	63
6 Commutative Algebra	65
6.1 Standard Problems (★)	65
6.2 Moderate Problems (★★)	66
6.3 Challenging Problems (★★★)	72
7 Representation Theory	73
7.1 Standard Problems (★)	73
7.2 Moderate Problems (★★)	74
7.3 Challenging Problems (★★★)	77

8 Homological Algebra	79
8.1 Standard Problems (★)	79
8.2 Moderate Problems (★★)	79
A Problems by Exam	85
Spring 2000	85
Fall 2000	85
Spring 2001	86
Fall 2001	86
Spring 2002	87
Winter 2002	87
Fall 2002	88
Winter 2003	88
Fall 2003	88
Spring 2004	89
Fall 2004	89
Winter 2005	90
Fall 2005	90
Spring 2006	90
Fall 2006	91
Spring 2007	91
Fall 2007	91
Spring 2008	92
Fall 2008	92
Spring 2009	93
Fall 2009	93
Spring 2010	94
Fall 2010	94
Spring 2011	95
Fall 2011	95
Spring 2012	95
Fall 2013	96
Spring 2013	96
Fall 2014	97
Spring 2015	97
Fall 2016	97
Spring 2016	98
Fall 2017	98
Spring 2017	99
Fall 2018	99
Spring 2018	100
Fall 2019	100
Spring 2019	100
Fall 2020	101
Spring 2020	101
Fall 2021	102
Spring 2021	102
Fall 2022	102

Spring 2022	103
Fall 2023	103
Spring 2023	104
Fall 2024	104
Spring 2024	104
Fall 2025	105
Spring 2025	105
Spring 2026	106
B Top Problems by Interest	107
Group Theory	107
Ring Theory	108
Module Theory	108
Field Theory & Galois Theory	108
Linear Algebra	109
Commutative Algebra	110
Representation Theory	110
Homological Algebra	110

Preface

This document collects all extractable problems from UCLA algebra qualifying examinations from Spring 2000 through Spring 2026. The exams from 2000–2011 were digitized via OCR and are included here; their mathematical notation may contain minor OCR artifacts.

Problems are organized into eight chapters by topic. Within each chapter, problems are sorted by difficulty:

- ★ — Standard: direct application of definitions or classical theorems.
- ★★ — Moderate: requires combining multiple ideas or moderate cleverness.
- ★★★ — Hard: requires significant insight, unusual constructions, or advanced techniques.

Within the same difficulty level, problems appear in chronological order. An index at the end lists all problems organized by exam.

Source: <https://ww3.math.ucla.edu/past-qualifying-exams/>

Chapter 1

Group Theory

Problems on groups, subgroups, normal subgroups, homomorphisms, Sylow theory, group actions, abelian groups, solvable and nilpotent groups.

1.1 Standard Problems (★)

Problem 1.1 Exam: Spring 2000, A1 | Difficulty: ★ | Interest: ◇

State a theorem which classifies (i.e. lists) all finite abelian groups up to isomorphism. This means that each finite abelian group should be isomorphic to exactly one group of your list. Use your classification to list abelian groups of order 24.

Problem 1.2 Exam: Fall 2000, A1 | Difficulty: ★ | Interest: ◇

Let D_{2n} be the dihedral group of order $2n$ with $n > 1$. Determine the number of subgroups of D_{2n} of index 2, and justify your answer.

Problem 1.3 Exam: Spring 2001, A1 | Difficulty: ★ | Interest: ◇

Determine a complete set of groups of order eight up to isomorphism and show that every group of order eight is isomorphic to one of these.

Problem 1.4 Exam: Fall 2001, G1 | Difficulty: ★ | Interest: ◇

Let G be a finite group whose center has index n . Show that every conjugacy class in G has at most n elements.

Problem 1.5 Exam: Spring 2012, #5 | Difficulty: ★ | Interest: ◇

Let G be a finite group, K a normal subgroup, and P a p -Sylow subgroup of G . Prove that $P \cap K$ is a p -Sylow subgroup of K .

Problem 1.6 Exam: Spring 2012, #10 | Difficulty: ★ | Interest: ◇

Determine the automorphism group of the symmetric group S_3 up to isomorphism.

Problem 1.7 Exam: Fall 2013, #1 | Difficulty: ★ | Interest: ◇

How many groups are there up to isomorphism of order pq where $p > q$ are prime integers?

Problem 1.8 Exam: Spring 2015, #1 | Difficulty: ★ | Interest: ◇

What are the coproducts in the category of groups?

Problem 1.9 Exam: Spring 2015, #8 | Difficulty: ★ | Interest: ◇◇◇

Let G be a finite group of order pq , where p and q are distinct primes. Show that

- (a) G has a normal subgroup distinct from 1 and G ;
- (b) if $p \not\equiv 1 \pmod{q}$ and $q \not\equiv 1 \pmod{p}$, then G is abelian.

Problem 1.10 Exam: Fall 2016, #1 | Difficulty: ★ | Interest: ◇◇◇

Let G be a group generated by a and b with only relation $a^2 = b^2 = 1$ (the group identity). Determine the group structure of G and justify your answer.

1.2 Moderate Problems (★★)

Problem 1.11 Exam: Spring 2000, A2 | Difficulty: ★★ | Interest: ◇◇

Let S_5 be the symmetric group on 5 letters. For each positive integer n , list the number of elements of S_5 of order n . Justify your answer.

Problem 1.12 Exam: Fall 2000, A2 | Difficulty: ★★ | Interest: ◇◇

A group of order a power of a prime p is called a p -group. Let G be a finite group. Prove that for any given prime p , there exists a unique normal subgroup N of G such that (i) G/N is a p -group and (ii) any homomorphism φ of G into a p -group is trivial on N (that is, $\varphi(N) = 1$).

Problem 1.13 Exam: Spring 2001, A2 | Difficulty: ★★ | Interest: ◇◇

A finite group G acts on itself by conjugation. Determine all possible G if this action yields precisely three orbits. Prove your result.

Problem 1.14 Exam: Fall 2001, G2 | Difficulty: ★★ | Interest: ◇◇

Let G be a subgroup of S_n , that acts transitively on the set $\{1, 2, \dots, n\}$. Let H be the stabilizer in G of an element $z \in \{1, 2, \dots, n\}$. Prove that $(\cap_{g \in G} gHg^{-1}) = \{e\}$.

Problem 1.15 Exam: Spring 2002, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Show that a group G of order $2m$, where m odd, has a normal subgroup of order m .

Problem 1.16 Exam: Spring 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

List, up to isomorphism, all finite abelian groups A satisfying the following two conditions: (i) A is a quotient of $\mathbb{Z}^n$, and (ii) A is annihilated by 18, i.e. $18a = e$ for all a in A . Your list should contain a representative of each isomorphism class exactly once. How many groups are there?

Problem 1.17 Exam: Winter 2002, A1 | Difficulty: ★★ | Interest: ◇◇

Let G be a free abelian group of rank n for a positive integer n (therefore $G \cong \mathbb{Z}^n$ as groups). (a) Prove for a given integer $m > 1$, there are only finitely many subgroups H of index m in G ; (b) Find a formula of the number of subgroups of G of index 3. Justify your answer.

Problem 1.18 Exam: Winter 2002, A2 | Difficulty: ★★ | Interest: ◇◇

Prove or disprove: there exists a finite abelian group G whose automorphism group has order 3.

Problem 1.19 Exam: Winter 2002, A8 | Difficulty: ★★ | Interest: ◇◇

3. Let S and G be p -groups (with $G \neq \{e\}$), and assume that S acts on G by automorphisms. Show that the fixed subgroup $G^* = \{g \in G \mid s(g) = g \text{ for all } s \in S\}$ is non-trivial (i.e., is not the trivial subgroup $\{e\}$).

Problem 1.20 Exam: Fall 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of order 108. Show that G has a normal subgroup of order 9 or 27.

Problem 1.21 Exam: Fall 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group and P a p -Sylow subgroup. Let $N_G(P)$ be the normalizer of P in G . Show that: a) P is the unique p -Sylow subgroup of $N_G(P)$ (Do not quote a theorem that this is true!) b) $N_G(P)$ is self normalizing in G Tee eee

Problem 1.22 Exam: Winter 2003, A1 | Difficulty: ★★ | Interest: ◇◇

List, up to isomorphism, all abelian groups A which satisfy the following three conditions: (i) A has 108 elements; (ii) A has an element of order 9; (iii) A has no element of order 24.

Problem 1.23 Exam: Winter 2003, A2 | Difficulty: ★★ | Interest: ◇◇

Let $N > 1$ be a positive integer. Show that a finitely generated group G has only finitely many subgroups of index at most N .

Problem 1.24 Exam: Winter 2003, A3 | Difficulty: ★★ | Interest: ◇◇

Let $N > 2$ be an integer. Show that a subgroup of index 2 in S_N is A_N . Here S_N and A_N are the symmetric and alternating groups for N , respectively. eo o 2

Problem 1.25 Exam: Fall 2003, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a finitely generated group, and $n > 1$ an integer. Show that G has at most a finite number of subgroups of index n .

Problem 1.26 Exam: Fall 2003, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group, K a normal subgroup, and P a p -Sylow subgroup of K for some prime p . Prove that $G = KNe@(\mathcal{P})$.

Problem 1.27 Exam: Fall 2003, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Suppose G is the free abelian group on generators z, y, z, w , considered as an additive group. Let $a = 2 - z + 4 + 2uw, b = x^2 - yt + u, c = 3xr - y - 2z + 5u, d = 2x - 2y + 4w$. If $H = (a, b, c, d)$, determine the structure of G/H .

Problem 1.28 Exam: Spring 2004, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

A group G is said to act transitively on a set S if for any element $s \in S$, then $S = Gs$. Suppose G is finite and that G acts transitively on S . Let $f(g)$ be the number of elements of S fixed by the action of $g \in G$ on S . Prove $|G| = \sum_{g \in G} f(g)$.

Problem 1.29 Exam: Spring 2004, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a finite group and H a subgroup of G . Let $n = (G : H)$ be the index of H in G . (a) Show that $(G : H)$ is a factor of $|G|$. (b) Suppose that the index $(G : H)$ is the minimal prime factor of the order of G . Show H is a normal subgroup.

Problem 1.30 Exam: Fall 2004, Problem 3 | Difficulty: ★★ | Interest: ◇◇

Linear Algebra

- Determine whether it is true in general that in $GL_n(\mathbb{C})$ every matrix is conjugate to its transpose.
- Determine the number of conjugacy classes in $GL_3(\mathbb{C})$ whose elements A satisfy the polynomial $X^2 - 2X + 1 = 0$.
- Let A be an n -by- n symmetric matrix with real entries and let, for $j \in [1, n]$, A_j be the submatrix consisting of the entries of A in the first j rows and columns of A . Show A is positive definite iff $\det(A_j) > 0$ for all $j \in [1, n]$.

Problem 1.31 Exam: Winter 2005, A1 | Difficulty: ★★ | Interest: ◇◇

- If G is a simple group that has subgroup of index n , prove that the order of G is a factor of $n!$.
- Prove that there is no simple nonabelian group of order p^*m with $e > 0$ for a prime $p > m$.

Problem 1.32 Exam: Winter 2005, A2 | Difficulty: ★★ | Interest: ◇◇

An additive abelian group is called divisible if multiplication by n for every positive integer n is a surjective endomorphism. (a) Show that if G is divisible, G/H is divisible for any

subgroup

Problem 1.33 Exam: Winter 2005, A3 | Difficulty: ★★ | Interest: ◇◇

Let G be a free abelian group of finite rank r . (a) Show that there are only finitely many homomorphisms of G into $\mathbb{Z}/n\mathbb{Z}$ for each positive integer n . (b) Find a formula of the number of surjective homomorphisms of G onto $\mathbb{Z}/p\mathbb{Z}$ for a prime p if $r = 2$.

Problem 1.34 Exam: Fall 2005, Sec.1 | Difficulty: ★★ | Interest: ◇◇◇

- (a) Let G be an abelian group generated by n elements. Prove that every subgroup of G can also be generated by n elements.
 (b) Let N be a normal subgroup of G . Prove that for a Sylow p -subgroup P of G , the intersection PMN is a Sylow p -subgroup of N .
 (c) Is there a nontrivial action of the alternating group A_n on a set of two elements?

Problem 1.35 Exam: Spring 2006, G1 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group. Let K be a normal subgroup of G and P a p -Sylow subgroup of K . Show that $G = KN_G(P)$.

Problem 1.36 Exam: Spring 2006, G2 | Difficulty: ★★ | Interest: ◇◇

(a) What is the order of $SL_2(\mathbb{F}_4)$? (b) Show there is an isomorphism from $SL_2(\mathbb{F}_4)$ to A_5 . Hint: Consider the action of $SL_2(\mathbb{F}_4)$ on the set of one dimensional subspaces of the vector space $(\mathbb{F}_4)^2$ of dimension two over $\mathbb{F}_4$.

Problem 1.37 Exam: Spring 2006, G3 | Difficulty: ★★ | Interest: ◇◇

Let G be a group of order 2000 and suppose that P and P' are two distinct Sylow 5 subgroups of G . Let $I = \langle P, P' \rangle$. (a) Prove that $|I| = 25$. (b) Show that the index of $N_G(I)$ is at most 2.

Problem 1.38 Exam: Fall 2006, G1 | Difficulty: ★★ | Interest: ◇◇

List all finite groups G whose automorphism group has prime order. Justify your answer.

Problem 1.39 Exam: Fall 2006, G2 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group and H be a non-normal subgroup of G of index $n > 1$. (a) Show that if $|H|$ is divisible by a prime $p \leq n$, then H cannot be a simple group; (b) Show that there is no simple group of order $504 = 2^3 \cdot 3^2 \cdot 7$. (Hint: Choose a good prime p , and let G act on the set of Sylow p -subgroups getting an embedding of G into a permutation group, and apply (a). You may use the fact that the alternating group A_n is simple if $n > 5$.)

Problem 1.40 Exam: Fall 2006, G3 | Difficulty: ★★ | Interest: ◇◇

Let $SL_2(\mathbb{Z}/p\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}/p\mathbb{Z}, ad - bc = 1 \right\}$, where p is an odd prime.
 (a) Prove that any subgroup of a cyclic group is cyclic.

- (b) Compute the order of G .
 (c) Prove that for any odd prime ℓ , the Sylow ℓ -subgroup of G is cyclic. (Hint: You may use the fact that the multiplicative group $F^\times$ of a finite field F is cyclic).

Problem 1.41 Exam: Spring 2007, Problem 1 | Difficulty: ★★ | Interest: ◇◇

Let G be a simple group containing an element of order 21. Prove that every proper subgroup of G has index at least 10.

Problem 1.42 Exam: Spring 2007, Problem 3 | Difficulty: ★★ | Interest: ◇◇

Let G be a group with cyclic automorphism group $\text{Aut}(G)$. Prove that G is abelian.

Problem 1.43 Exam: Fall 2007, G1 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field of characteristic p and let G be a subgroup of order p^* of the group $\text{GL}(N, F)$ of invertible $N \times N$ matrices with entries in F . Show that there is a non-zero vector v in F^N such that $gv = v$ for every $g \in G$.

Problem 1.44 Exam: Fall 2007, G2 | Difficulty: ★★ | Interest: ◇◇

Let M be the submodule of $\mathbb{Z}^3$ generated by elements $(0, 3, 2)$, $(6, 48, 24)$ and $(6, 24, 12)$. Describe the quotient group $\mathbb{Z}^3/M$ by giving a product of cyclic groups to which $\mathbb{Z}^3/M$ is isomorphic.

Problem 1.45 Exam: Fall 2007, G3 | Difficulty: ★★ | Interest: ◇◇

A group G acts doubly transitively on a set X if for each pair (x_1, x_2) and (y_1, y_2) , with $x_1 \neq x_2$ and $y_1 \neq y_2$ of pairs of points of X , there exists a $g \in G$ such that $gx_1 = y_1$ and $gx_2 = y_2$. Show that if a finite G acts non-trivially and doubly transitively on X , then the stabilizer S_x of any point x in X is a maximal proper subgroup of G . (Here a subgroup M of G is maximal proper if M is not equal to G and if, for any subgroup H of G which contains M , either $H = M$ or $H = G$ holds.)

Problem 1.46 Exam: Spring 2008, PROBLEM G1 | Difficulty: ★★ | Interest: ◇◇

Let p be a prime number. Show that a subgroup G of S_n , which contains an element of order p and which contains a transposition must be the whole of S_n .

Problem 1.47 Exam: Spring 2008, PROBLEM G2 | Difficulty: ★★ | Interest: ◇◇

Let $G = D_{2n}$ be the dihedral group of order $2n$ where $n > 3$. Prove that $\text{Aut}(G)$ is isomorphic to the group of 2×2 matrices of the form $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$ where $a \in (\mathbb{Z}/n\mathbb{Z})^\times$, $b \in \mathbb{Z}/n\mathbb{Z}$.

Problem 1.48 Exam: Spring 2008, PROBLEM G3 | Difficulty: ★★ | Interest: ◇◇

Let K be a normal subgroup of a finite group G . Let p be a prime. Let N_p be the number of p -Sylow subgroups of G and $N_p(K)$, the number in K . (a) Show that $N_p = [G : N_p(K)]$

where P is any p -SSG of G and $N_G(P)$ is the normalizer of P in G . (b) Prove that N_j divides N_p .

Problem 1.49 Exam: Fall 2008, G1 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of order g and $Z[G] \subset Q[G]$ be the group algebras of G with integer and rational coefficients, respectively. Let $eZ[G] = \{ea \in Q[G] \mid a \in Z[G]\}$ for $e = g^{-1} \sum_{g \in G} g$. Define $G' = eZ[G]/(Z[G]N_e Z[G])$. Prove that G' is a group of order g . Find a necessary and sufficient condition to have $G' \cong G$ as groups, and justify your answer.

Problem 1.50 Exam: Fall 2008, G2 | Difficulty: ★★ | Interest: ◇◇

Prove or disprove:

- (a) the group $GL_2(\mathbb{Q})$ of 2×2 matrices with rational coefficients has finite cyclic subgroups of order bigger than any given positive integer N .
- (b) the group $GL_2(\mathbb{R})$ of 2×2 matrices with real coefficients has finite cyclic subgroups of order bigger than any given positive integer N .

Problem 1.51 Exam: Fall 2008, G3 | Difficulty: ★★ | Interest: ◇◇

Let G be an additive abelian group such that multiplication by n : $rt \mapsto nx$ is surjective for all positive integers n . Let $G[n] = \{x \in G \mid nx = 0\}$ and p be a prime.

- (a) Prove that for a given integer $m > 1$, there are only finitely many subgroups H of order p^m in G if $G[p]$ is finite;
- (b) Find a formula of the number of subgroups of G of order p if the order of $G[p]$ is p^k , and justify your answer.

Problem 1.52 Exam: Spring 2009, G1 | Difficulty: ★★ | Interest: ◇◇

Let N be a normal subgroup of a finite group G and P a Sylow p -subgroup of G . Prove that PMN is a Sylow p -subgroup of N .

Problem 1.53 Exam: Spring 2009, G2 | Difficulty: ★★ | Interest: ◇◇

Let A be an abelian group generated by n elements. Prove that any subgroup of A can be generated by n elements.

Problem 1.54 Exam: Spring 2009, G3 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group and $H \subset G$ a subgroup of index n . Suppose that $HN \cap H \neq 0$ for any elements $x, y \in G$. Prove that $|G| > n^2 - n$. (Hint: consider an action of G on $(G/H) \times (G/H)$.)

Problem 1.55 Exam: Fall 2009, (Gr 1) | Difficulty: ★★ | Interest: ◇◇

Suppose A is an abelian group that is generated by n elements (or fewer). Show that any subgroup of A also can be generated by n elements (or fewer).

Problem 1.56 Exam: Fall 2009, (Gr 2) | Difficulty: ★★ | Interest: ◇◇

Let $p < q$ be primes, $n > 0$ an integer and G a group of order pq^n . Show that G is solvable.

Problem 1.57 Exam: Spring 2010, Problem 3 | Difficulty: ★★ | Interest: ◇◇

Determine, using the structure theory of abelian groups or otherwise, all finitely generated abelian groups A such that the group $\text{Aut}(A)$ of automorphisms of A is finite. State clearly any basic theorems that you use. Determine the order of the automorphism group of $A = \mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$.

Problem 1.58 Exam: Spring 2010, Problem 4 | Difficulty: ★★ | Interest: ◇◇◇

Show that if S_n denotes the symmetric group of degree n , and σ is an automorphism of S_n , and $\tau \in S_n$ is a transposition, then $\sigma(\tau)$ is also a transposition. By studying the action of σ on transpositions, show that every automorphism of S_n is inner. (Remark: this result holds for all n , except $n=6$.)

Problem 1.59 Exam: Fall 2010, Problem 4 | Difficulty: ★★ | Interest: ◇◇◇

1. Show from first principles (i.e. without using the classification theorem) that a subgroup of a finitely generated abelian group is finitely generated.

Problem 1.60 Exam: Fall 2010, Problem 2 | Difficulty: ★★ | Interest: ◇◇

Let $M \subset \mathbb{Z}^3$ be the subgroup generated by elements $(13, 9, 2)$, $(29, 21, 5)$, and $(2, 2, 2)$. Determine the isomorphism class of the quotient group

Problem 1.61 Exam: Spring 2011, Problem 1 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a group of order n . Show that there are two subgroups H and H' of the symmetric group S_n , both isomorphic to G such that $h_1 h_2 = a h_3$ for all $a_1 \in A_1$ and $h_3 \in A_0$.

Problem 1.62 Exam: Spring 2011, Problem 2 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a finitely generated group. Show that G contains only finitely many subgroups of any fixed finite index.

Problem 1.63 Exam: Spring 2011, Problem 10 | Difficulty: ★★ | Interest: ◇◇

Show that there is a (covariant) functor from the category of groups to the category of sets taking a group G to the set of all subgroups of G . Determine whether this functor is representable. —

Problem 1.64 Exam: Fall 2011, Problem 2 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a non-trivial finite group and p a prime. If every subgroup H of G has index divisible by p , prove that the center of G has order divisible by p .

Problem 1.65 Exam: Fall 2011, Problem 9 | Difficulty: ★★ | Interest: ◇◇

(We denote by FG the group algebra of G .)

(a) Is there a group G with CG isomorphic to $C \times C \times M_2(C)$? (b) Is there a group G with QG isomorphic to $Q \times Q \times M_3(Q)$?

Problem 1.66 Exam: Fall 2013, #2 | Difficulty: ★★ | Interest: ◇◇

Show that there are up to isomorphism exactly two nonabelian groups G of order 8. Prove that each of them has an irreducible complex representation of dimension 2.

Problem 1.67 Exam: Spring 2013, #1 | Difficulty: ★★ | Interest: ◇◇

Let G be a free abelian group of rank r , so $G \cong \mathbb{Z}^r$ as groups. Show that G has only finitely many subgroups of a given finite index n .

Problem 1.68 Exam: Fall 2014, #6 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group and let p be the smallest prime number dividing the order of G . Assume G has a normal subgroup H of order p . Show that H is contained in the center of G .

Problem 1.69 Exam: Spring 2015, #2 | Difficulty: ★★ | Interest: ◇◇

Let $\mathcal{C}$ be the category of groups and $\mathcal{C}'$ be its full subcategory with objects the abelian groups. Let $F : \mathcal{C}' \rightarrow \mathcal{C}$ be the inclusion functor. Determine the left adjoint of F and show that F has no right adjoint.

Problem 1.70 Exam: Spring 2015, #10 | Difficulty: ★★ | Interest: ◇◇

Let E , M , and F be finite abelian groups and consider group homomorphisms $E \xrightarrow{f} M \xrightarrow{g} F$. Assume g is injective. Show that $|\text{Coker}(g \circ f)| = |\text{Coker}(g)| \cdot |\text{Coker}(f)|$, where $|X|$ denotes the order of a finite set X .

Problem 1.71 Exam: Spring 2016, #9 | Difficulty: ★★ | Interest: ◇◇

Show that if G is a finite group acting transitively on a set X with at least two elements, then there exists $g \in G$ which fixes no point of X .

Problem 1.72 Exam: Fall 2017, #1 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group, p a prime number, and S a Sylow p -subgroup of G . Let $N = \{g \in G \mid gSg^{-1} = S\}$. Let X and Y be two subsets of $Z(S)$ (the center of S) such that there is $g \in G$ with $gXg^{-1} = Y$. Show that there exists $n \in N$ such that $nXn^{-1} = gYg^{-1}$ for all $x \in X$.

Problem 1.73 Exam: Fall 2017, #2 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of order a power of a prime p . Let $\Phi(G)$ be the subgroup of G generated by elements of the form g^p for $g \in G$ and $ghg^{-1}h^{-1}$ for $g, h \in G$. Show that $\Phi(G)$ is the intersection of the maximal proper subgroups of G (the Frattini subgroup).

Problem 1.74 Exam: Fall 2018, #1 | Difficulty: ★★ | Interest: ◇◇◇

Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ be the quaternion group of order 8.

- (1) Show that every non-trivial subgroup of Q_8 contains -1 .
- (2) Show that Q_8 does not embed in the symmetric group S_7 (as a subgroup).

Problem 1.75 Exam: Fall 2018, #2 | Difficulty: ★★ | Interest: ◇◇

Let G be a finitely generated group having a subgroup of finite index $n > 1$. Show that G has finitely many subgroups of index n and has a proper characteristic subgroup (preserved by all automorphisms) of finite index.

Problem 1.76 Exam: Spring 2018, #9 | Difficulty: ★★ | Interest: ◇◇

Show that there is no simple group of order 616.

Problem 1.77 Exam: Fall 2019, #1 | Difficulty: ★★ | Interest: ◇◇◇

Show that every group of order 315 is the direct product of a group of order 5 with a semidirect product of a normal subgroup of order 7 and a subgroup of order 9. How many such isomorphism classes are there?

Problem 1.78 Exam: Fall 2019, #6 | Difficulty: ★★ | Interest: ◇◇

Classify all finite subgroups of $GL(2, \mathbb{R})$ up to conjugacy.

Problem 1.79 Exam: Spring 2019, #1 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a finite solvable group and $1 \neq N \subset G$ be a minimal normal subgroup. Prove that there exists a prime p such that N is either cyclic of order p or a direct product of cyclic groups of order p .

Problem 1.80 Exam: Fall 2020, #1 | Difficulty: ★★ | Interest: ◇◇

Let $p < q < r$ be primes and G a group of order pqr . Prove that G is not simple and, in fact, has a normal Sylow r -subgroup.

Problem 1.81 Exam: Fall 2020, #2 | Difficulty: ★★ | Interest: ◇◇

Show that groups of order $231 = 3 \cdot 7 \cdot 11$ are semidirect products and show that there are exactly two such groups up to isomorphism.

Problem 1.82 Exam: Spring 2020, #1 | Difficulty: ★★ | Interest: ◇◇

Let G be a group defined by $G = \langle a, b \mid a^2 = b^2 = 1 \rangle$. Determine the order of all non-trivial finite quotient groups.

Problem 1.83 Exam: Spring 2020, #6 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field with at least 3 elements. Show that $SL_2(F)$ has order divisible by 12.

Problem 1.84 Exam: Spring 2020, #7 | Difficulty: ★★ | Interest: ◇◇

Let G be a p -group and $1 \neq N \trianglelefteq G$ be a non-trivial normal subgroup.

- (a) Show that N contains a non-trivial element of the center $Z(G)$ of G .
- (b) Give an example where $Z(N) \not\subseteq Z(G)$.

1.3 Challenging Problems (★★★)

Problem 1.85 Exam: Spring 2000, A3 | Difficulty: ★★★ | Interest: ◇◇

Let F_4 be the field with 4 elements. Let $G = SL(2, F_4)$ be the group of 2 by 2 invertible matrices with entries in F_4 . What is the order of G ? Show, by analysing the action of G on the lines containing the origin in $(F_4)^*$, that G is a simple group. (Hint: how many lines containing the origin are there?)

Problem 1.86 Exam: Fall 2000, A3 | Difficulty: ★★★ | Interest: ◇◇◇

Let G be a finite group of order n . Suppose that G has a unique subgroup of order d for each positive divisor d of n . Prove that G is a cyclic group.

Problem 1.87 Exam: Spring 2001, B3 | Difficulty: ★★★ | Interest: ◇◇

Let $f_1(z_1, \dots, z_n)$, $f_2(z_1, \dots, z_n)$, $f_3(z_1, \dots, z_n)$ be m polynomials in

Problem 1.88 Exam: Fall 2001, G8 | Difficulty: ★★★ | Interest: ◇◇◇

Let G be the group of matrices of the form $\begin{pmatrix} a & b & 0 \\ 0 & 1 & 0 \end{pmatrix}$ where $a \in (Z/p)^*$ and $b \in Z/p$. Describe all normal subgroups of G . Hint: find a convenient normal subgroup of order p .

Problem 1.89 Exam: Fall 2004, Problem 1 | Difficulty: ★★★ | Interest: ◇◇◇

Groups

- (a) Is there a finite group G such that $G/Z(G)$ has 143 elements? ($Z(G)$ is the center of G .)
- (b) Prove that every group of order 30 has a subgroup of order 15.
- (c) Find all finite groups that have exactly three conjugacy classes.

Problem 1.90 Exam: Fall 2014, #1 | Difficulty: ★★★ | Interest: ◇◇◇

Let G be a finite group. Let $\mathbb{Z}[G]$ be the group algebra of G with augmentation ideal $\mathfrak{a}$. Show that $\mathfrak{a}/\mathfrak{a}^2 \cong G/G'$ as abelian groups, where G' is the derived subgroup of G .

Problem 1.91 Exam: Fall 2014, #7 | Difficulty: ★★★ | Interest: ◇◇

Let G be a finite group and P a Sylow 2-subgroup of G . Assume P is cyclic, generated by an element x . Show that the signature of the permutation of G given by $g \mapsto xg$ is -1 . Deduce that G has a non-trivial quotient of order 2.

Problem 1.92 Exam: Spring 2017, #3 | Difficulty: ★★★ | Interest: ◇◇

Find the number of subgroups of index 3 in the free group $F_2 = \langle u, v \rangle$ on two generators. Justify your answer.

Problem 1.93 Exam: Spring 2020, #2 | Difficulty: ★★★ | Interest: ◇◇

Let G be a finite group of order $n > 1$ and consider its group algebra $\mathbb{Z}[G]$ embedded in $\mathbb{Q}[G]$. Let $A = \mathbb{Z}[G]/\mathfrak{a}$ for the ideal $\mathfrak{a}$ generated by $g - 1$ for all $g \in G$.

- (a) Prove that the algebra $\mathbb{Q}[G]$ is the product of $\mathbb{Q}$ and $\mathbb{Q} \cdot \mathfrak{a}$, where $\mathbb{Q} \cdot \mathfrak{a}$ is the $\mathbb{Q}$ -span of $\mathfrak{a}$ in $\mathbb{Q}[G]$.
- (b) Let B be the projected image of $\mathbb{Z}[G]$ in $\mathbb{Q} \cdot \mathfrak{a}$. Prove that $A \otimes_{\mathbb{Z}[G]} B \cong G$ as groups if and only if G is a cyclic group.

Chapter 2

Ring Theory

Problems on rings, ideals, Jacobson radical, semisimple rings, division rings, polynomial rings, PIDs, UFDs.

2.1 Standard Problems (★)

Problem 2.1 Exam: Spring 2000, B1 | Difficulty: ★ | Interest: ◇

List, up to isomorphism, all commutative rings with 4 elements. Prove your answer.

Problem 2.2 Exam: Fall 2000, B1 | Difficulty: ★ | Interest: ◇

Let M be a module over a commutative ring A . If every increasing (resp. decreasing) sequence of A -submodules of M terminates after finite steps, the A -module M is called noetherian (resp. artinian). (a) Prove that the $\mathbb{Z}$ -module $\mathbb{Z}$ is noetherian and non-artinian. (b) Prove that the $\mathbb{Z}$ -module $\mathbb{Q}$, $(\mathbb{p}^{-n}\mathbb{Z}/\mathbb{Z})$ is artinian and non-noetherian.

Problem 2.3 Exam: Fall 2013, #9 | Difficulty: ★ | Interest: ◇

Prove that if $ab = 1$ in a semisimple ring, then $ba = 1$.

Problem 2.4 Exam: Spring 2013, #5 | Difficulty: ★ | Interest: ◇

Let R be a commutative local ring with unique maximal ideal M .

- (a) Show that if $x \in M$ then $1 - x$ is invertible.
- (b) Show that if R is Noetherian and I is an ideal such that $I^2 = I$, then $I = 0$.

Problem 2.5 Exam: Fall 2014, #8 | Difficulty: ★ | Interest: ◇

Let A be a ring. Assume there is an infinite chain of left ideals $I_0 \subset I_1 \subset \cdots \subset A$ with $I_i \neq I_{i+1}$ for $i \geq 0$. Show that A has a left ideal that is not finitely generated as a left A -module.

Problem 2.6 Exam: Spring 2016, #1 | Difficulty: ★ | Interest: ◇

(a) Give an example of a unique factorization domain A that is not a PID. (You need not show A is a UFD, but please show your example is not a PID.) (b) Let R be a UFD. Let $\mathfrak{p}$ be a prime ideal such that $0 \neq \mathfrak{p}$ and there is no prime ideal strictly between 0 and $\mathfrak{p}$. Show that $\mathfrak{p}$ is principal.

2.2 Moderate Problems (★★)

Problem 2.7 Exam: Spring 2000, B2 | Difficulty: ★★ | Interest: ◇◇

Let p be a prime number. Show that a free $\mathbb{Z}$ module of rank 2 has $p+1$ submodules of index p .

Problem 2.8 Exam: Fall 2000, B2 | Difficulty: ★★ | Interest: ◇◇

Let A be a commutative ring with identity. Suppose that $a \in A$ is not nilpotent (that is, $a^n \neq 0$ for all $n > 0$). (a) Prove that there exists a prime ideal $\mathfrak{p} \subset A$ such that $a \notin \mathfrak{p}$; (b) Give an example of a ring A and a non-nilpotent $a \in A$ such that a is contained in $\mathfrak{m}$ for all maximal ideals $\mathfrak{m} \subset A$. Justify your example.

Problem 2.9 Exam: Fall 2001, R2 | Difficulty: ★★ | Interest: ◇◇

(a) Give an example of prime ideal in a commutative ring that is not maximal. (b) Let R be a commutative ring with identity. Suppose for every element $x \in R$ there exists an integer $n = n(x) > 1$ such that $x^n = z$. Show that every prime ideal in R is maximal.

Problem 2.10 Exam: Fall 2001, R3 | Difficulty: ★★ | Interest: ◇◇

8. (20 points) Let R be a ring. (a) Prove that if a is a nilpotent element in a ring R with identity, then the element $1 + a$ is invertible. In the next two parts, let $f(X) = a_0 + a_1X + \dots + a_nX^n$ be a polynomial in $R[X]$ of degree n , that is, $a_n \neq 0$. (b) Show that if R is an integral domain, then $f(X)$ is invertible in $R[X]$ if and only if $a_0 = 0$. (c) Show that if R is a commutative ring, $f(X)$ is invertible in $R[X]$ if and only if all a_i are invertible and a_i are nilpotent in R for every $i > 1$.

Problem 2.11 Exam: Spring 2002, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Let R be a ring and A and B be two non-isomorphic simple, left R -modules (a left-module is simple if it has no proper submodules, i.e., submodules other than $\{0\}$ and itself). Show that the only proper submodules of $M = A \oplus B$ are $\{(a, 0) : a \in A\}$ and $\{(0, b) : b \in B\}$.

Problem 2.12 Exam: Spring 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative local ring, that is, R has a unique maximal ideal M .

(i) Show that if x lies in M , then $1 - x$ is invertible.

(ii) Show that if R is Noetherian and I is an ideal satisfying $I^2 = J$, then $I = 0$. Hint: consider a minimal set of generators for I .

Problem 2.13 Exam: Spring 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let F_2 be the field with 2 elements and let $R = F_2[X]$. List, up to isomorphism, all R -modules of order 8. :

Problem 2.14 Exam: Winter 2002, B1 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and A be a commutative F -algebra. Suppose A is of finite dimension as a vector space of F . (a) Prove all prime ideals of A are maximal. Hint: consider maps $R/P \rightarrow R/P$ (P prime) of the form $x \mapsto az$ with a in R . (b) Prove that there are only finitely many maximal ideals of A .

Problem 2.15 Exam: Winter 2002, B2 | Difficulty: ★★ | Interest: ◇◇

Let $A = M_n(F)$ be the ring of $n \times n$ matrices with entries in an infinite field F for $n > 1$. Prove the following facts: (a) There are only 2 two-sided ideals of A ; (b) There are infinitely many maximal left ideals of A . Hint: show that $Az = Ay$ ($x, y \in A$) if and only if $\text{Ker}(z) = \text{Ker}(y)$.

Problem 2.16 Exam: Winter 2002, B3 | Difficulty: ★★ | Interest: ◇◇

Let F_2 be the field with 2 elements and $A = F_2[T, 4]$ for an indeterminate T . Prove the following facts: (a) The group of units in A is generated by T . (b) There are infinitely many distinct ring endomorphisms of A . (c) The ring automorphism group $\text{Aut}(A)$ is of order 2. e 1

Problem 2.17 Exam: Fall 2002, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative ring with 1, and let $S = R[x]$ be the polynomial ring in one variable. Suppose M is a maximal ideal of S . Prove that M cannot consist entirely of 0-divisors. Hint: You may want to distinguish the cases $c \in M$ or $c \notin M$

Problem 2.18 Exam: Fall 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative ring with 1, and suppose I and J are ideals of R so that: $1 + IJ = R$. Show that: (i) $IJ = I \cap J$ (ii) $R/IJ = R/(I \cap J)$

Problem 2.19 Exam: Fall 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative ring with 1, and let $S = R[x]$ be the polynomial ring in one variable. Let $f \in S$. If f is a unit of S (that is, f is invertible in S), show that f has the form $f = u + g$ or $f = u + g$ where u is a unit in R and $g \in S$ is a nilpotent element without constant term.

Problem 2.20 Exam: Winter 2003, B1 | Difficulty: ★★ | Interest: ◇◇

Give an example of two integral domains A and B which contain a field F such that $A \otimes_F B$ is not an integral domain. Justify your answer. Hint: Take A to be the field of rational functions $F(X)$ for the field F , with p elements.

Problem 2.21 Exam: Winter 2003, B2 | Difficulty: ★★ | Interest: ◇◇

Let F be the finite field of q elements, and put $F' = F$, and $K = F'$. Write $\sigma : K \rightarrow K$ for the field automorphism given by $x^\sigma = 2x$. Let $B = \{ \begin{pmatrix} d & 0 \\ 0 & d \end{pmatrix} \in M_2(K) \}$ for a given $d \in F'^\times$. Prove the following three facts: (a) B is a subalgebra of dimension 4 over F inside the F' -algebra of 2×2 matrices over K . (b) B is a division algebra if and only if there exists no $c \in K$ such that $d = cc^\sigma$. (c) B cannot be a division algebra.

Problem 2.22 Exam: Winter 2003, B3 | Difficulty: ★★ | Interest: ◇◇

Let A be a discrete valuation ring with maximal ideal M , and define $B = \{ (a, b) \in A \times A \mid a \equiv b \pmod{M} \}$. Prove the following facts: (a) B has only one maximal ideal; (b) B has exactly two non-maximal prime ideals.

Problem 2.23 Exam: Fall 2003, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let R be a ring, not necessarily commutative, and M a Noetherian left R -module. Suppose $f: M \rightarrow M$ is a surjective R -module map from M to M . Prove that f is an isomorphism.

Problem 2.24 Exam: Spring 2004, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative noetherian ring with unity 1 and $f: R \rightarrow R$ a surjective ring homomorphism, i.e. $f(R) = R$. Show f is an isomorphism.

Problem 2.25 Exam: Spring 2004, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let R be the ring $\mathbb{Q}[z]$ and let M be the submodule of R^n generated by the elements $(1 - 2x, -z)$ and $(1 - x, x - z)$. According to the theory of modules over principal ideal domains, R^n/M is a direct sum of cyclic R modules of the form $R/P(x)$ for monic polynomials $P(x)$. Find such a direct sum decomposition explicitly in this case.

Problem 2.26 Exam: Spring 2004, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇◇

Suppose we are given a collection of polynomials in r variables with rational coefficients: $f_1, \dots, f_n \in \mathbb{Q}[T_1, \dots, T_r]$. We define the complex algebraic set $V \subset \mathbb{C}^r$ by $V = \{ (a_1, \dots, a_r) \mid f_i(a_1, \dots, a_r) = 0 \text{ for all } i \text{ from } 1 \text{ to } n \}$. Suppose V is not empty. Show that there is a finite extension K of $\mathbb{Q}$ and a point $(a_1, \dots, a_r) \in V$ with all $a_i \in K$.

Problem 2.27 Exam: Fall 2004, Problem 2 | Difficulty: ★★ | Interest: ◇◇

Rings

(a) Let X be a finite set and let A be the ring of all functions from X to the field $\mathbb{R}$ of real numbers. Prove that an ideal M of A is maximal if and only if there is an element $x \in X$ such that

$$M = \{ f \in A \mid f(x) = 0 \}.$$

(b) Describe all $n \in \mathbb{Z}$ such that the ring $\mathbb{Z}/n\mathbb{Z}$ has no idempotents other than 0 and 1.

(c) A (non-commutative) ring R is called local if for every $a \in R$ either a or $1 - a$ is invertible. Prove that non-invertible elements of a local ring form a (two-sided) ideal.

Problem 2.28 Exam: Winter 2005, C1 | Difficulty: ★★ | Interest: ◇◇

Let R be an integral domain. If $\mathfrak{m}$ is a maximal ideal in R , view the localization $R_{\mathfrak{m}} := S^{-1}R$, with $S = R \setminus \mathfrak{m}$, in the quotient field of R . Show that $R = \bigcap_{\mathfrak{m} \in \text{Max}(R)} R_{\mathfrak{m}}$.

Problem 2.29 Exam: Winter 2005, C2 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative Artinian ring. Show that there are only finitely many prime ideals in R and every one of them is maximal.

Problem 2.30 Exam: Winter 2005, C3 | Difficulty: ★★ | Interest: ◇◇

Let $R \subset A \subset B$ be commutative rings. Suppose that FR is noetherian and B is a finitely generated R -algebra. Suppose that as an A -module B is finitely generated. Show that A is a finitely generated R -algebra.

Problem 2.31 Exam: Fall 2005, Sec.2 | Difficulty: ★★ | Interest: ◇◇

- (a) Let I and J be ideals of a commutative ring R with unit such that $I + J = R$. Prove that $1 - J = I \cap J$.
- (b) Prove that the factor ring $R[z, y]/(y^2 - z^2)R[z, y]$ is not a P.I.D.
- (c) Let z, y, z, t be elements of a (non-commutative) ring R such that $xz = yt = 1$, $ct = yz = 0$ and $zz + ty = 1$. Prove that the left R -modules R and $R @ R$ are isomorphic.

Problem 2.32 Exam: Spring 2006, R1 | Difficulty: ★★ | Interest: ◇

Suppose D is an integral domain and suppose that $D[z]$ is a principal ideal domain. Show D is a field.

Problem 2.33 Exam: Spring 2006, R2 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative Noetherian ring with unit, and suppose M is a finitely generated R module. Suppose $f : M \rightarrow M$ is an R module homomorphism which is onto. Show that f is an isomorphism.

Problem 2.34 Exam: Spring 2006, R3 | Difficulty: ★★ | Interest: ◇◇

- Let R be a commutative ring with unit and $\mathfrak{m}$ a maximal ideal of R .
- (a) Suppose $J, \dots, I$, are ideals of R and that where $I, \dots, J$, is the product of the ideals. Show for some k .
 - (b) Suppose that R satisfies the descending chain condition (dcc) on ideals, i.e. every strictly decreasing sequence of ideals is finite. Show FR has only a finite number of maximal ideals. You may use part (a), but not theorems on the structure of rings satisfying the dcc.

Problem 2.35 Exam: Fall 2006, R2 | Difficulty: ★★ | Interest: ◇◇

Let R be a noetherian domain. A non-zero element z in R is called a prime element if (z) is a prime ideal. Prove all of the following:

- (a) Every nonzero non-unit in R is a product of irreducible elements. (b) Every nonzero ideal $J \neq R$ in R contains a (finite) product of non-zero prime ideals. (c) If every nonzero prime ideal in R contains a prime element then every irreducible element in R is a prime element. [You may not use theorems about UFD's]

Problem 2.36 Exam: Fall 2006, R3 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative ring and M a finitely generated R -module. Suppose there exists a positive integer n and a surjective R -module homomorphism $\gamma : M \rightarrow R^n$. Show that $\ker \gamma$ is also a finitely generated R -module.

Problem 2.37 Exam: Spring 2007, Problem 1 | Difficulty: ★★ | Interest: ◇◇

Let D be a division ring (a ring with identity in which every non-zero element is invertible). Let $R = \text{Mat}_n(D)$ be the ring of $n \times n$ matrices with entries from D . Prove that R has no two-sided ideals other than R itself and $\{0\}$.

Problem 2.38 Exam: Spring 2007, Problem 2 | Difficulty: ★★ | Interest: ◇◇

Let $R = \text{End}(V)$ be the ring of all linear endomorphisms of an infinite dimension complex vector space V with countable basis $\{e_1, e_2, \dots\}$. Prove that R and $R \otimes R$ are isomorphic as left R -modules.

Problem 2.39 Exam: Spring 2007, Problem 3 | Difficulty: ★★ | Interest: ◇◇

(a) Give a description of all maximal ideals of the ring $\mathbb{C}[z, y]$. Justify your description. You may use the Nullstellensatz. (b) Let $M = (x^2 - y, y^2 - 5)$ be an ideal in $R = \mathbb{Q}[z, y]$. Prove that M is a maximal ideal.

a

Problem 2.40 Exam: Fall 2007, R1 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and A be a commutative F -algebra. Suppose A is of finite dimension as a vector space over F . (a) Prove that if A is a domain, A is a field. (b) Prove that even if A is not a domain, there are only finitely many prime ideals of A .

Problem 2.41 Exam: Fall 2007, R2 | Difficulty: ★★ | Interest: ◇◇

Let A be a commutative ring with identity, and write V for the set of all prime ideals of A . Put $D(z) = \{P \in V \mid z \notin P\}$ for $z \in A$. Prove

- (a) $D(a) = D(a^n)$ for integers $n > 0$; (b) $V = D(a) \cup D(b) \cup D(c)$ if $a^3 + 6b^2 + c$ is invertible in A .

Problem 2.42 Exam: Spring 2008, PROBLEM R1 | Difficulty: ★★ | Interest: ◇◇

Let D be an associative ring with unit having no zero divisors. Assume that the center of D contains a field k such that $\dim_k(D) < \infty$. Prove that D is a division algebra (i.e., every non-zero element is invertible).

Problem 2.43 Exam: Spring 2008, PROBLEM R2 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of order $|G| > 1$. The rational group ring $\mathbb{Q}[G]$ of G is the $\mathbb{Q}$ -algebra consisting of all finite linear combinations $\sum a_g g$ where $a_g \in \mathbb{Q}$. Multiplication in $\mathbb{Q}[G]$ is defined by extending the group multiplication linearly. (a) Show that $\mathbb{Q}[G]$ has a non-trivial idempotent: $e \in \mathbb{Q}[G]$, $e \neq 0, 1$ with $e^2 = e$. Hint: reduce to the case of a cyclic group $G = \langle x : x^n = 1 \rangle$. (b) Show that $\mathbb{Q}[G]$ contains an invertible element u that is non-trivial, i.e., not of the form $u = ag$ where $a \in \mathbb{Q}$ and $g \in G$. (FYI: The Kadison-Kaplansky Conjecture claims that for G without torsion, the group algebra $\mathbb{Q}[G]$ contains no non-trivial idempotent. This conjecture is open in general.)

Problem 2.44 Exam: Spring 2008, PROBLEM R3 | Difficulty: ★★ | Interest: ◇◇

Let R be a Noetherian ring and J any ideal of R . Prove that there exist prime ideals $P_1, \dots, P_n$ of R such that $J \subseteq \bigcap_{i=1}^n P_i$. Hint: Show that if J is any non-prime ideal, then there exist $a, b \notin J$ such that $(J + a)(J + b) \subseteq J$. Then use the Noetherian property.

Problem 2.45 Exam: Fall 2008, R1 | Difficulty: ★★ | Interest: ◇◇

Let $A = M_n(F)$ be the ring of $n \times n$ matrices with entries in a field F .

- (a) Prove that any left ideal of A is principal of the form Az ;
- (b) How many left ideals of A if $n = 2$ and F is a finite field? (Give a simple formula of the number of maximal left ideals of A , and justify your answer.)

Problem 2.46 Exam: Fall 2008, R2 | Difficulty: ★★ | Interest: ◇◇

Let A be a domain and $B = A[T]$ for an indeterminate T . Prove that the ring automorphism group $\text{Aut}(B/A)$ of B inducing the identity on A is finite if and only if the group of invertible elements of A is finite.

Problem 2.47 Exam: Fall 2008, R3 | Difficulty: ★★ | Interest: ◇◇

8. Consider the covariant functor $F: A \mapsto A^*$ from the category $\mathbf{ALG}$ of commutative rings with identity to the category of sets. Here A^* is the group of invertible elements of A . Give an explicit form of F .

Problem 2.48 Exam: Spring 2009, R1 | Difficulty: ★★ | Interest: ◇◇

Show that the ring $\mathbb{Z}[2i]$ consisting of all complex numbers $a + 2bi$ with $a, b \in \mathbb{Z}$ is not a PID.

Problem 2.49 Exam: Spring 2009, R2 | Difficulty: ★★ | Interest: ◇◇

Let $M_n(F)$ be the matrix ring of $n \times n$ matrices over a field F . Suppose that there is a subring of $M_n(F)$ isomorphic to $M_m(F)$ for some m . Prove that m divides n .

Problem 2.50 Exam: Spring 2009, R3 | Difficulty: ★★ | Interest: ◇◇

Two polynomials $f, g \in \text{Rit}$ over a commutative ring R are called coprime over R if f and g generate the unit ideal in $R[\]$. Let $f, g \in \mathbb{Z}[t]$ be two polynomials such that f and g are coprime over $\mathbb{Q}$ and the residues of f and g in $(\mathbb{Z}/p\mathbb{Z})[t]$ are coprime for every prime p . integer p . Prove that f and g are coprime over $\mathbb{Z}$.

Problem 2.51 Exam: Fall 2011, Problem 4 | Difficulty: ★★ | Interest: ◇◇

Let p be a prime. Prove that the nilradical of the ring $\mathbb{F}_p[X]_{(p, X^2)}$ is a principal ideal.

Problem 2.52 Exam: Spring 2012, #4 | Difficulty: ★★ | Interest: ◇◇

Let R be the subring of $M_3(\mathbb{R})$ consisting of all matrices (a_{ij}) with $a_{31} = a_{32} = 0$. Determine the Jacobson radical of R .

Problem 2.53 Exam: Fall 2013, #10 | Difficulty: ★★ | Interest: ◇◇

Let A be the functor from the category of groups to the category of (unital) rings taking a group G to the group ring $\mathbb{Z}[G]$ of all finite formal sums $\sum_{g \in G} a_g g$ with $a_g \in \mathbb{Z}$. Prove that A has a right adjoint functor.

Problem 2.54 Exam: Spring 2013, #6 | Difficulty: ★★ | Interest: ◇◇

Let D be a division ring of characteristic 0. Assume that D has dimension 2 as a $\mathbb{Q}$ -vector space. Show that D is commutative.

Problem 2.55 Exam: Spring 2013, #9 | Difficulty: ★★ | Interest: ◇◇

Let D be an associative ring without zero divisors, and assume the center of D is a field over which D is a finite-dimensional vector space. Prove that D is a division algebra.

Problem 2.56 Exam: Fall 2014, #10 | Difficulty: ★★ | Interest: ◇◇

Let n be a positive integer. Let A_n be the $\mathbb{Q}$ -algebra generated by elements $x_1, \dots, x_n, y_1, \dots, y_n$ with relations $x_i x_j = x_j x_i$, $y_i y_j = y_j y_i$, and $y_i x_j - x_j y_i = \delta_{ij}$ for $1 \leq i, j \leq n$. Show that there is a representation of A_n on the vector space $\mathbb{Q}[t_1, \dots, t_n]$ where x_i acts by multiplication by t_i and y_i acts as $\frac{\partial}{\partial t_i}$.

Problem 2.57 Exam: Spring 2015, #3 | Difficulty: ★★ | Interest: ◇◇

Let R be a ring. Show that R is a division ring if and only if all R -modules are free.

Problem 2.58 Exam: Spring 2015, #7 | Difficulty: ★★ | Interest: ◇◇

Determine the ring endomorphisms of $\mathbb{F}_2[t, t^{-1}]$, where t is an indeterminate.

Problem 2.59 Exam: Spring 2015, #9 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of order p^n for a prime p . Show that the group ring $\mathbb{F}_p[G]$ over the finite field $\mathbb{F}_p$ has a unique maximal two-sided ideal.

Problem 2.60 Exam: Fall 2016, #10 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and A a simple subalgebra of a finite-dimensional F -algebra B . Prove that $\dim_F(A)$ divides $\dim_F(B)$.

Problem 2.61 Exam: Spring 2016, #3 | Difficulty: ★★ | Interest: ◇◇

Let R be a ring which is left artinian. Suppose that R is a domain, meaning that $1 \neq 0$ in R and $ab = 0$ implies $a = 0$ or $b = 0$. Show that R is a division ring.

Problem 2.62 Exam: Spring 2017, #6 | Difficulty: ★★ | Interest: ◇◇

Prove the following generalization of Nakayama's Lemma to non-commutative rings. Let R be a ring with 1 (not necessarily commutative) and suppose that $J \subset R$ is a left ideal contained in every maximal left ideal of R . If M is a finitely generated left R -module such that $JM = M$, prove that $M = 0$.

Problem 2.63 Exam: Spring 2018, #3 | Difficulty: ★★ | Interest: ◇◇

Let $\mathbb{Z}^n$ ($n > 1$) be made of column vectors with integer coefficients. Prove that for every non-zero left ideal I of $M_n(\mathbb{Z})$, the subgroup $I\mathbb{Z}^n$ (generated by products αv with $\alpha \in I$ and $v \in \mathbb{Z}^n$) has finite index in $\mathbb{Z}^n$.

Problem 2.64 Exam: Spring 2019, #3 | Difficulty: ★★ | Interest: ◇◇

Let $d > 2$ be a square-free integer. Show that the integer 2 in $\mathbb{Z}[\sqrt{-d}]$ is irreducible but the ideal (2) in $\mathbb{Z}[\sqrt{-d}]$ is not a prime ideal.

Problem 2.65 Exam: Spring 2019, #7 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and let R be the ring of 3×3 matrices over F with $(3, 1)$ and $(3, 2)$ entries equal to 0:

$$R = \begin{pmatrix} F & F & F \\ F & F & F \\ 0 & 0 & F \end{pmatrix}.$$

- Determine the Jacobson radical J of R .
- Is J a minimal left (respectively, right) ideal?

2.3 Challenging Problems (★★★)

Problem 2.66 Exam: Spring 2000, B3 | Difficulty: ★★★ | Interest: ◇◇

Let R be a commutative noetherian ring in which each ideal I is principal and satisfies $J(I) \subseteq I$. Show that R is isomorphic to a finite product of fields.

Problem 2.67 Exam: Fall 2000, B3 | Difficulty: ★★★ | Interest: ◇◇◇

Let A be a commutative noetherian ring with identity $1 \neq 0$. Write $X(a)$ for the set of prime ideals of A containing a given ideal a . Suppose that $X(0) = X(a) \cup X(b)$ and $X(a) \cap X(b) = \emptyset$ for two ideals a and b . Prove the following facts: (a) $A = a + b$; (b) $a \cap b = ab$; (c) The ideal ab consists of nilpotent elements. Hint: You may use the assertion of B2 (a); (d) There exists a positive integer n such that A is isomorphic to the product ring $(A/a^n) \times (A/b^n)$.

Problem 2.68 Exam: Fall 2007, R3 | Difficulty: ★★★ | Interest: ◇◇

Determine all isomorphism classes of modules over the polynomial ring $F_2[X]$ which are of dimension 2 over F_2 , and justify your answer. Here F_2 is a field of two elements.

Problem 2.69 Exam: Fall 2014, #4 | Difficulty: ★★★ | Interest: ◇◇◇

Let D be a 9-dimensional central division algebra over $\mathbb{Q}$ and $K \subset D$ be a field extension of $\mathbb{Q}$ of degree > 1 . Show that $K \otimes_{\mathbb{Q}} K$ is not a field and deduce that $D \otimes_{\mathbb{Q}} K$ is no longer a division algebra.

Problem 2.70 Exam: Fall 2016, #2 | Difficulty: ★★★ | Interest: ◇◇◇

Let K be a semisimple quadratic extension over $\mathbb{Q}$ and consider the regular representation $\rho : K \rightarrow M_2(\mathbb{Q})$. Compute the index of $\rho(K^\times)$ in the normalizer of $\rho(K^\times)$ in $GL_2(\mathbb{Q})$, and justify your answer.

Problem 2.71 Exam: Spring 2016, #5 | Difficulty: ★★★ | Interest: ◇◇◇

Let A be the ring $\mathbb{C}\langle u, v \rangle / (uv - vu - 1)$, the quotient of the free associative algebra on two generators by the given two-sided ideal.

- (a) Show that every nonzero A -module M has infinite dimension as a complex vector space.
- (b) Let M be an A -module with a nonzero element y such that $uy = 0$. Show that the elements $y, vy, v^2y, \dots$ are $\mathbb{C}$ -linearly independent in M .

Problem 2.72 Exam: Fall 2017, #3 | Difficulty: ★★★ | Interest: ◇◇◇

Let k be a field and A a finite-dimensional k -algebra. Denote by $J(A)$ the Jacobson radical of A . Let $t : A \rightarrow k$ be a morphism of k -vector spaces such that $t(ab) = t(ba)$ for all $a, b \in A$. Assume $\ker(t)$ contains no nonzero left ideal. Let M be the set of elements a in A such that $t(xa) = 0$ for all $x \in J(A)$. Show that M is the largest semisimple left A -submodule of A .

Problem 2.73 Exam: Spring 2018, #4 | Difficulty: ★★★ | Interest: ◇◇◇

Let p be a prime number, and let D be a central simple division algebra of dimension p^2 over a field k . Pick $\alpha \in D$ not in the center and write K for the subfield of D generated by α . Prove that $D \otimes_k K \cong M_p(K)$ (the $p \times p$ matrix algebra with entries in K).

Problem 2.74 Exam: Fall 2020, #3 | Difficulty: ★★★ | Interest: ◇◇◇

A ring R (commutative or non-commutative) is called a domain if $ab = 0$ in R implies $a = 0$ or $b = 0$. Suppose that R is a domain such that $M_n(R)$, the ring of $n \times n$ matrices over R , is a semisimple ring. Prove that R is a division ring.

Chapter 3

Module Theory

Problems on modules over rings, free and projective modules, exact sequences, tensor products, structure theorems.

3.1 Standard Problems (★)

Problem 3.1 Exam: Spring 2019, #4 | Difficulty: ★ | Interest: ◇

Let R be a commutative local ring and P a finitely generated projective R -module. Prove that P is R -free.

Problem 3.2 Exam: Fall 2020, #4 | Difficulty: ★ | Interest: ◇

Let M be a left R -module. Show that M is a projective R -module if and only if there exist $m_i \in M$ and R -homomorphisms $f_i : M \rightarrow R$ for each $i \in I$ such that:

- (a) if $m \in M$, then $f_i(m) = 0$ for all but finitely many $i \in I$;
- (b) if $m \in M$, then $m = \sum_{i \in I} f_i(m)m_i$.

3.2 Moderate Problems (★★)

Problem 3.3 Exam: Spring 2001, B2 | Difficulty: ★★ | Interest: ◇◇

Let $R = R\{z, y\}$. Find a finitely generated R -module M that is not a direct sum of cyclic R -modules, and prove that it is not.

Problem 3.4 Exam: Spring 2012, #3 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative local ring, and P a finitely generated projective R -module. Show that P is a free module.

Problem 3.5 Exam: Spring 2013, #4 | Difficulty: ★★ | Interest: ◇◇

Let $\mathbb{F}_2$ be the field with 2 elements and let $R = \mathbb{F}_2[x]$. List, up to isomorphism, all R -modules with 8 elements.

Problem 3.6 Exam: Fall 2014, #9 | Difficulty: ★★ | Interest: ◇◇

Let A be a ring and let $i, j \in A$ such that $i^2 = i$ and $j^2 = j$. Show that the left A -modules Ai and Aj are isomorphic if and only if there are $a, b \in A$ such that $i = ab$ and $j = ba$.

Problem 3.7 Exam: Spring 2015, #4 | Difficulty: ★★ | Interest: ◇◇

Let $M = \mathbb{Z}[1/p]/\mathbb{Z}$ and $N = \mathbb{Q}/\mathbb{Z}$, where $\mathbb{Z}[1/p] \subset \mathbb{Q}$ is the subring generated by $1/p$ for a prime p . Show that

- (a) M is an artinian module but not a noetherian module;
- (b) N is neither noetherian nor artinian.

Problem 3.8 Exam: Fall 2016, #3 | Difficulty: ★★ | Interest: ◇◇

Let A be an integral domain with field of fractions F . For an A -ideal $\mathfrak{a}$, prove that $\mathfrak{a}$ is a finitely generated projective A -ideal if and only if there exists an A -submodule $\mathfrak{b}$ of F such that $\mathfrak{a}\mathfrak{b} = A$.

Problem 3.9 Exam: Spring 2017, #8 | Difficulty: ★★ | Interest: ◇◇

Let M be an abelian group (written additively). Prove that there is a functor F from the opposite of the category of rings to the category of sets taking a ring R to the set of all left R -module structures on M . Is the functor F representable?

Problem 3.10 Exam: Spring 2017, #9 | Difficulty: ★★ | Interest: ◇◇

Let R be a ring. Prove that if the left free R -modules R^n and R^m are isomorphic for some positive integers n and m , then R^n and R^m are isomorphic as right R -modules.

Problem 3.11 Exam: Fall 2018, #9 | Difficulty: ★★ | Interest: ◇◇

Let $f : M \rightarrow N$ and $g : N \rightarrow M$ be two R -linear homomorphisms of R -modules such that $\text{id}_M - gf$ is invertible. Show that $\text{id}_N - fg$ is invertible as well and give a formula for its inverse.

Problem 3.12 Exam: Spring 2018, #5 | Difficulty: ★★ | Interest: ◇◇

Let $\mathcal{C}$ be a category. A morphism $f : A \rightarrow B$ in $\mathcal{C}$ is called an *epimorphism* if for any two morphisms $g, h : B \rightarrow X$, $g \circ f = h \circ f$ implies $g = h$. Let **ALG** be the category of $\mathbb{Z}$ -algebras, and let **MOD** be the category of $\mathbb{Z}$ -modules.

- (a) Prove that in **MOD**, $f : M \rightarrow N$ is an epimorphism if and only if f is a surjection.
- (b) In **ALG**, does the equivalence of epimorphism and surjection hold? If yes, prove it; if no, give a counterexample.

Problem 3.13 Exam: Fall 2019, #4 | Difficulty: ★★ | Interest: ◇◇

Find all isomorphism classes of simple (i.e., irreducible) left modules over the ring $M_n(\mathbb{Z})$ of $n \times n$ matrices with $\mathbb{Z}$ -entries, for $n \geq 1$.

Problem 3.14 Exam: Fall 2019, #5 | Difficulty: ★★ | Interest: ◇◇

Let R be a nonzero commutative ring. Consider the functor t_B from the category of R -modules to itself given by taking the (right) tensor product with an R -module B .

- (a) Prove that t_B commutes with colimits.
- (b) Construct an R -module B (for each R) such that t_B does not commute with limits.

Problem 3.15 Exam: Fall 2019, #8 | Difficulty: ★★ | Interest: ◇◇

Let M be a finitely generated module over an integral domain R . Show that there is a nonzero element $u \in R$ such that the localization $M[1/u]$ is a free module over $R[1/u]$.

Problem 3.16 Exam: Spring 2019, #2 | Difficulty: ★★ | Interest: ◇◇

An additive group Q (abelian group written additively) is called *divisible* if any equation $nx = y$ with $0 \neq n \in \mathbb{Z}$, $y \in Q$ has a solution $x \in Q$. Let Q be a divisible group and A a subgroup of an abelian group B . Give a complete proof that every group homomorphism $A \rightarrow Q$ can be extended to a group homomorphism $B \rightarrow Q$.

Problem 3.17 Exam: Spring 2020, #9 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative ring and $S \subset R$ a multiplicative subset. Construct a natural transformation (in either direction) between the functors $\text{Hom}_{S^{-1}R}(S^{-1}M, S^{-1}N)$ and $S^{-1}\text{Hom}_R(M, N)$, considered as functors of R -modules M and N , and prove it is an isomorphism if M is finitely presented.

Problem 3.18 Exam: Spring 2020, #10 | Difficulty: ★★ | Interest: ◇◇◇

Let R be a commutative ring and M a left R -module. Let $f : M \rightarrow M$ be a surjective R -linear endomorphism.

- (a) Suppose that M is finitely generated. Show that f is an isomorphism and that f^{-1} can be described as a polynomial in f . (Hint: Let $R[X]$ act on M via f .)
- (b) Show that this fails if M is not finitely generated.

3.3 Challenging Problems (★★★)

Problem 3.19 Exam: Fall 2017, #5 | Difficulty: ★★★ | Interest: ◇◇◇

Let A be a ring and M an A -module that is a finite direct sum of simple A -modules. Let $f \in \text{End}_{\mathbb{Z}}(M)$. Assume $f \circ g = g \circ f$ for all $g \in \text{End}_A(M)$. Consider a positive integer n .

- Show that the map $f_n : M^n \rightarrow M^n$ defined by $f_n(m_1, \dots, m_n) = (f(m_1), \dots, f(m_n))$ commutes with all elements of $\text{End}_A(M^n)$.
- Deduce that given any family $(m_1, \dots, m_n) \in M^n$, there exists $a \in A$ such that $(f(m_1), \dots, f(m_n)) = (am_1, \dots, am_n)$.

Problem 3.20 Exam: Fall 2017, #9 | Difficulty: ★★★ | Interest: ◇◇◇

Let R be a commutative ring, and let A , B , and C be R -modules. Suppose that A is finitely presented over R and C is flat over R . Show that

$$\text{Hom}_R(A, B \otimes_R C) \cong \text{Hom}_R(A, B) \otimes_R C.$$

Problem 3.21 Exam: Fall 2018, #10 | Difficulty: ★★★ | Interest: ◇◇◇

Consider the real algebra $A = \mathbb{R}[x, y] = \mathbb{R}[X, Y]/(X^2 + Y^2 - 1)$ where x and y are the classes of X and Y . Let $M = A(1+x) + Ay$ be the ideal generated by $1+x$ and y .

- Show that there is an A -linear isomorphism $A^2 \xrightarrow{\sim} M \oplus M$ mapping the canonical basis to $(1+x, y)$ and $(-y, 1+x)$.
- Show that there is an A -linear isomorphism $A \xrightarrow{\sim} M \otimes_A M$ mapping 1 to $((1+x) \otimes (1+x)) + (y \otimes y)$.

Problem 3.22 Exam: Spring 2020, #8 | Difficulty: ★★★ | Interest: ◇◇◇

Let R be a ring.

- Show that an R -module X is indecomposable if $\text{End}_R(X)$ is local.
- Suppose that every finitely generated R -module M is isomorphic to $X_1 \oplus \dots \oplus X_m$ with all $\text{End}_R(X_i)$ local. Show that such a decomposition is unique (Krull–Schmidt theorem).
- Give an example of an isomorphism $X_1 \oplus X_2 \cong Y_1 \oplus Y_2$ with $\text{End}(X_i)$ and $\text{End}(Y_i)$ local, that is not the direct sum of any isomorphisms $X_i \cong Y_i$, even up to renumbering.

Chapter 4

Field Theory & Galois Theory

Problems on field extensions, splitting fields, Galois groups, separability, cyclotomic fields, finite fields.

4.1 Standard Problems (★)

Problem 4.1 Exam: Fall 2000, C1 | Difficulty: ★ | Interest: ◇

Find the minimal polynomial of $\sqrt{2} + \sqrt[3]{3}$ over the field of rational numbers $\mathbb{Q}$.

Problem 4.2 Exam: Spring 2001, C1 | Difficulty: ★ | Interest: ◇

Let F be a prime field, i.e., the rationals or a field with p elements. Prove that an algebraic closure of F has infinite degree over F . [Hint: You may want to do the two cases separately.]

Problem 4.3 Exam: Fall 2001, F1 | Difficulty: ★ | Interest: ◇

Let $f(z) = z^2 - 2z - 2$. (a) Show that $f(z)$ is irreducible over $\mathbb{Q}$. (b) Let α be a complex root of $f(z)$. Express $6!$ as a polynomial in α with coefficients in $\mathbb{Q}$.

Problem 4.4 Exam: Fall 2016, #6 | Difficulty: ★ | Interest: ◇

Let F be a field of characteristic $p > 0$. Prove that for every $a \in F$, the polynomial $x^p - a$ is either irreducible or splits into a product of linear factors.

Problem 4.5 Exam: Spring 2017, #10 | Difficulty: ★ | Interest: ◇

Let K/F be a (finite) Galois field extension with $G = \text{Gal}(K/F)$ and let $H \subset G$ be a subgroup. Determine in terms of H and G the group $\text{Gal}(K^H/F)$ of all field automorphisms of K^H over F .

Problem 4.6 Exam: Spring 2019, #6 | Difficulty: ★ | Interest: ◇

Let F be a field of characteristic $p > 0$ and $a \in F^\times$. Prove that if the polynomial $f = X^p - a$ has no root in F , then f is irreducible over F .

4.2 Moderate Problems (★★)

Problem 4.7 Exam: Spring 2000, C1 | Difficulty: ★★ | Interest: ◇◇

Let $a = 1 + \sqrt[3]{2} + 374$.

- (a) Find the degree of a over $\mathbb{Q}$. Justify your answer.
- (b) Find a normal closure of $\mathbb{Q}(a)/\mathbb{Q}$. Justify your answer.

Problem 4.8 Exam: Spring 2000, C2 | Difficulty: ★★ | Interest: ◇◇

Let g be a power of a prime integer, $n \in \mathbb{N}$. Let k be the least positive integer such that $g^k \equiv 1 \pmod{n}$. Prove that the finite field $\mathbb{F}_n$ is a splitting field of the polynomial $X^n - 1$ over $\mathbb{F}_g$.

Problem 4.9 Exam: Spring 2000, C3 | Difficulty: ★★ | Interest: ◇◇

Find a subfield F in the field of rational functions $\mathbb{C}(X)$ such that $\mathbb{C}(X)/F$ is a Galois extension with the Galois group isomorphic to the symmetric group S_n . (Hint: Consider automorphisms of $\mathbb{C}(X)$ given by $X \mapsto aX + b$, $a, b \in \mathbb{C}$).

Problem 4.10 Exam: Fall 2000, C2 | Difficulty: ★★ | Interest: ◇◇

Let α be a primitive 16-th root of unity over the field F . Determine the dimension $[\mathbb{F}_p(\alpha) : \mathbb{F}_p]$ when F is:

- a) The field of 9 elements,
- b) The field of 7 elements,
- c) The field of 17 elements,
- d) What can you say in the case $p = 2$?

Problem 4.11 Exam: Spring 2001, C2 | Difficulty: ★★ | Interest: ◇◇

Let $f \in \mathbb{Q}[z]$ be a polynomial of degree three. Let $K = \mathbb{Q}(\alpha)$ be a splitting field of f . Determine all the possible Galois groups of $K/\mathbb{Q}$, prove these are all such, and give explicit examples of K , i.e., determine a α or f .

Problem 4.12 Exam: Fall 2001, F2 | Difficulty: ★★ | Interest: ◇◇

Let $f(z) = z^n + nx + 2$ where n is an integer. Determine the (infinitely many) values of n for which f is irreducible over $\mathbb{Q}$.

Problem 4.13 Exam: Winter 2002, C2 | Difficulty: ★★ | Interest: ◇◇

A field extension $K/\mathbb{Q}$ is called biquadratic if it has degree 4 and if $K = \mathbb{Q}(\sqrt{a}, \sqrt{b})$ for some $a, b \in \mathbb{Q}$. (a) Show that a biquadratic extension is normal with Galois group $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and list all sub-extensions. (b) Prove that if $K/\mathbb{Q}$ is a normal extension of degree 4 with $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ then $K/\mathbb{Q}$ is biquadratic.

Problem 4.14 Exam: Winter 2002, C3 | Difficulty: ★★ | Interest: ◇◇

Let K be a finite extension of the field F with no proper intermediate subfields. (a) If K/F is normal, show that the degree $[K:F]$ is a prime. (b) Give an example to show that $[K:F]$ need not be prime if K/F is not normal, and justify your answer.

Problem 4.15 Exam: Fall 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let p be a prime integer such that $p \equiv 2$ or $3 \pmod{5}$. Prove that the polynomial $14X^{44} + X^{74} + X^{34} + 4$ is irreducible over $\mathbb{Z}/p\mathbb{Z}$.

Problem 4.16 Exam: Winter 2003, C1 | Difficulty: ★★ | Interest: ◇◇

Let F_q be the finite field of q elements. Answer the following questions:

(a) List all subfields of F_q for a prime p . Justify your answer. (b) Find a formula for the number of monic irreducible polynomials of degree 6 in $F_q[X]$. Justify your answer.

Problem 4.17 Exam: Winter 2003, C2 | Difficulty: ★★ | Interest: ◇◇

Let K/F be a quadratic extension of fields and M/F be a Galois extension over F containing K such that $\text{Gal}(M/K)$ is a cyclic group of odd prime order p . Answer the following two questions: (a) Determine the possible groups $\text{Gal}(M/F)$ up to isomorphisms, and justify your answer. (b) Find the number of intermediate fields L between F and M with $[L:F] = p$. Justify your answer.

Problem 4.18 Exam: Winter 2003, C3 | Difficulty: ★★ | Interest: ◇◇

Find the degree of the splitting field F of $X^3 - 3$ over the following fields:

(a) $\mathbb{Q}(\sqrt{-3})$ ($\mathbb{Q}$: the field of rational numbers); (b) $\mathbb{F}_7$, the field with 7 elements; (c) $\mathbb{F}_5$, the field with 5 elements, and justify your answer.

Problem 4.19 Exam: Fall 2003, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let $f(X)$ be a polynomial of degree $n > 0$ over a field F . (a) Prove that there is a field homomorphism $\alpha: F(X) \rightarrow F(X)$ such that $\alpha(X) = f(X)$. (b) Let L be the image of α . Prove that the field extension $F(X)/L$ is finite and find its degree. (c) Find the minimal polynomial of X over L .

Problem 4.20 Exam: Fall 2003, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let p be a prime integer. Suppose that the degree of every finite extension of a field F is divisible by p . Prove that the degree of every finite extension of F is a power of p .

Problem 4.21 Exam: Spring 2004, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Let K be a finite extension of $\mathbb{Q}$ obtained by adjoining to $\mathbb{Q}$ a root of $f(z) = z^6 + 3$.

(a) Show that K contains a primitive 6-th root of unity.
 (b) Show that K is a Galois extension of $\mathbb{Q}$.

(c) Determine the number of fields F' of degree 3 over $\mathbb{Q}$ with $F \subset K$.

Problem 4.22 Exam: Spring 2004, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Suppose that $f(x)$ is a polynomial in $\mathbb{Q}\{z\}$ of degree $d > 1$ with d roots $\xi_1, \dots, \xi_n \in \mathbb{C}$. If $x^2 = az$, for $a \in \mathbb{Q}$ different from -1 , prove that $f(z)$ is reducible.

Problem 4.23 Exam: Spring 2004, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇◇

Let K be a field and L a finite extension of K . Consider the set A of all elements $\alpha \in L$ with the property that $K[\alpha]$ is a Galois extension of K with an abelian Galois group $\text{Gal}(K[\alpha]/K)$. Show that A is a subfield of L containing K .

Problem 4.24 Exam: Fall 2004, Problem 4 | Difficulty: ★★ | Interest: ◇◇

Fields (a) Let a be an integer and let p be a prime. Show that if a is not a p -th power, then $X^p - a$ is irreducible over $\mathbb{Q}$.

(b) Show that if K and L are finite separable extensions of F' with K Galois over F' , such that $KL = L$, then $[KL:F] = [L:F][K:F]$. Show that if neither K nor L are Galois over F' , then this fact need not be true.

(c) By using several quadratic extensions of the rational function field in two variables $F = \mathbb{F}_2(X, Y)$ where $\mathbb{F}_2$ is the field with 2 elements, give an example of a field extension of finite degree of F that possesses infinitely many intermediate fields.

Problem 4.25 Exam: Winter 2005, D1 | Difficulty: ★★ | Interest: ◇◇

Show that the identity map is the only field automorphism of the real numbers. Show this is not true of the complex numbers.

Problem 4.26 Exam: Winter 2005, D3 | Difficulty: ★★ | Interest: ◇◇

Let K/F be a finite extension of finite fields. Prove that the norm map $N_r: K \rightarrow F$ is surjective.

Problem 4.27 Exam: Spring 2006, F2 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field. (a) Show that more than half the elements of F' are squares. (b) Show that every element of F' is the sum of two squares.

Problem 4.28 Exam: Spring 2006, F3 | Difficulty: ★★ | Interest: ◇◇

Let K be a finite extension of the field F with no proper intermediate fields. (a) If K/F is normal, show $[K:F]$ must be prime. (b) Give an example to show that $[K:F]$ need not be prime if K/F is not normal, explaining why your example works.

Problem 4.29 Exam: Fall 2006, F1 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field of positive characteristic p . Show that the unit group $F^\times$ of F is a cyclic group and that F is a Galois extension of $\mathbb{Z}/p\mathbb{Z}$.

Problem 4.30 Exam: Fall 2006, F3 | Difficulty: ★★ | Interest: ◇◇

Let $f(x)$ be an irreducible polynomial over F' and K/F' a normal extension. Show that $f(x)$ factors into irreducible polynomials, over K all of the same degree.

Problem 4.31 Exam: Spring 2007, Problem 1 | Difficulty: ★★ | Interest: ◇◇

Let $F = \mathbb{Q}(\zeta_5)$ where $\zeta_5 = e^{2\pi i/5}$ and let E/F be a Cyclic Galois extension of degree 5. Prove that there exists $a \in F$ such that $E = F(\sqrt[5]{a})$. Hint: find $a \in E$ such that $\sigma(a) = \zeta_5 a$, where σ is a generator of the Galois group $\text{Gal}(E/F)$.

Problem 4.32 Exam: Spring 2007, Problem 2 | Difficulty: ★★ | Interest: ◇◇

Let $K = \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{3})$.

- Prove that K has only one subfield $F \subset K$ such that $[F : \mathbb{Q}] = 2$.
- Find all subfields of K .
- Find an element $u \in K$ such that $K = \mathbb{Q}(u)$.
- Describe all elements $u \in K$ such that $K = \mathbb{Q}(u)$.

Problem 4.33 Exam: Spring 2007, Problem 3 | Difficulty: ★★ | Interest: ◇◇

Let $F = \mathbb{Z}/3\mathbb{Z}$. First explain why $\mathbb{F}_3[z]/(z^2 - 2)$ is isomorphic to $\mathbb{F}_3\{2z\}/(x^2 - 2x - 1)$. Then find an explicit isomorphism:

Problem 4.34 Exam: Fall 2007, F1 | Difficulty: ★★ | Interest: ◇◇

Let F be a field. Show that the unit group $F^\times$ of F is finitely generated if and only if F is finite.

Problem 4.35 Exam: Fall 2007, F3 | Difficulty: ★★ | Interest: ◇◇

Let $f(z)$ be an irreducible polynomial over the field F' and let K/F' be a finite extension.

- Define what it means for the extension K/F' to be normal.
- Show that if K is normal over F' , then, in $K[X]$, $f(z)$ factors into a product of irreducible polynomials of the same degree.
- Show by example that this result does not hold for K not normal.

Problem 4.36 Exam: Spring 2008, PROBLEM F2 | Difficulty: ★★ | Interest: ◇◇

Let $\zeta_n = \exp(2\pi i/n)$ be a primitive n th root of unity. Let $F = \mathbb{Q}(\zeta_n)$. Set $d_n = [F : \mathbb{Q}]$ (a) Let $n = 6$. Find an irreducible polynomial of degree d_6 in $\mathbb{Q}[z]$ whose roots generate F . (b) Let $n = 12$. Find an irreducible polynomial of degree d_{12} in $\mathbb{Q}[z]$ whose roots generate F .

Problem 4.37 Exam: Spring 2008, PROBLEM F3 | Difficulty: ★★ | Interest: ◇◇

Let E/F be a finite, separable extension of fields. Prove that there exists $a \in E$ such that $E = F(a)$. State clearly any theorems you use in the proof.

Problem 4.38 Exam: Fall 2008, F1 | Difficulty: ★★ | Interest: ◇◇

Let L/F be a cubic (of degree 3) field extension of characteristic zero. Prove that there is an element $a \in F$ and a cubic field extension L_p of the field $F_y = \mathbb{Q}(a)$ such that L is the composite FL_p of F and L_p over F_0 .

Problem 4.39 Exam: Fall 2008, F2 | Difficulty: ★★ | Interest: ◇◇

A field extension L/F is said to be balanced if every field homomorphism $L \rightarrow ZL$ over F is an isomorphism.

(a) Prove that every algebraic (possibly infinite) field extension is balanced; (b) Give an example of a balanced non-algebraic field extension.

Problem 4.40 Exam: Fall 2008, F3 | Difficulty: ★★ | Interest: ◇◇

Let p be a prime integer and F' a field such that the degree of every nontrivial finite field extension of F is divisible by p . Prove that for any finite field extension L/F , there exists a tower of field extensions $F = F_0 \subset F_1 \subset \dots \subset F_n = L$ such that $[F_{i+1} : F_i] = p$ for any $i = 0, \dots, n-1$.

Problem 4.41 Exam: Spring 2009, F2 | Difficulty: ★★ | Interest: ◇◇

Let a be transcendental over $\mathbb{Q}$. What is the minimal polynomial of a over $\mathbb{Q}(s)$?

Problem 4.42 Exam: Spring 2009, F3 | Difficulty: ★★ | Interest: ◇◇

(A) Which roots of unity are contained in quadratic extensions of $\mathbb{Q}$, and which extensions are these? (B) If $K/\mathbb{Q}$ is any field extension that contains a primitive n 'th root of unity, and n is odd, then prove K contains a primitive $2n$ 'th root of unity.

Problem 4.43 Exam: Fall 2009, (F1) | Difficulty: ★★ | Interest: ◇◇

Prove that the multiplicative group $F^\times$ of a field F is a cyclic group if and only if F is a finite field.

Problem 4.44 Exam: Fall 2009, (F2) | Difficulty: ★★ | Interest: ◇◇

Let $k = \mathbb{F}_2(t, s)$ be the field of fractions of two variable polynomial ring $\mathbb{F}_2[t, s]$, where $\mathbb{F}_2$ is the field with 2 elements. Write α , for a root of $T^2 + 7T + a = 0$ for $a \in k$ in an algebraic closure of k . An intermediate field M between K and k for a field extension K/k is a subfield in K containing k .

(a) How many intermediate fields between k and $k(\alpha, \beta)$? (b) How many intermediate fields between k and $k(t, V/s)$? Justify all your answers.

Problem 4.45 Exam: Spring 2010, Problem 11 | Difficulty: ★★ | Interest: ◇◇

Let F be a field of characteristic zero containing the p -th roots of unity for p a prime. Show that the cyclic extensions of degree p of F in any algebraic closure F' of F are in one to one correspondence with the subgroups of order p of $F^*/(F^*)^p$.

Problem 4.46 Exam: Spring 2011, Problem 3 | Difficulty: ★★ | Interest: ◇◇

Let R be a noetherian domain. Show that every nonzero nonunit in R is a product of irreducible elements and that R is a UFD if and only if every nonzero nonunit in R is a product of prime elements, where an element is prime if it generates a nonzero prime ideal.

4. Prove that every prime ideal in $\mathbb{Z}[t]$ can be generated by two elements.

5. Let F be a field having no nontrivial field extensions of odd degree and $F/\mathbb{Q}$ a finite field extension. Show if F has no field extensions of degree two, then F is perfect and F is algebraically closed.

Problem 4.47 Exam: Spring 2011, Problem 6 | Difficulty: ★★ | Interest: ◇◇

Prove that over a finite field there are irreducible polynomials of any positive degree.

Problem 4.48 Exam: Fall 2011, Problem 1 | Difficulty: ★★ | Interest: ◇◇

For a finite field F , prove that the order of the group $SL_2(F)$, of 2×2 matrices with determinant 1, is divisible by 6.

Problem 4.49 Exam: Fall 2011, Problem 5 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field and $\bar{F}$ be an algebraic closure of F . Let K be a subfield of $\bar{F}$ generated by all roots of unity over F . Show that any simple K -algebra of finite dimension over K is isomorphic to the matrix algebra $M_n(K)$ for a positive integer n .

Problem 4.50 Exam: Fall 2011, Problem 10 | Difficulty: ★★ | Interest: ◇◇

Let K/k be an extension of finite fields. Show that the norm $N_{K/k} : K \rightarrow k$ is surjective.

Problem 4.51 Exam: Spring 2012, #6 | Difficulty: ★★ | Interest: ◇◇

Determine the Galois group of the polynomial $X^8 + 16$ over $\mathbb{Q}$.

Problem 4.52 Exam: Spring 2012, #7 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field with p^n elements (p prime). Find the number of elements of F that can be written in the form $a^p - a$ for some $a \in F$.

Problem 4.53 Exam: Fall 2013, #3 | Difficulty: ★★ | Interest: ◇◇◇

For a positive integer n , let $\Phi_n(X)$ be the n -th cyclotomic polynomial. If a is an integer and p a prime not dividing n such that p divides $\Phi_n(a)$, show that the order of $a \bmod p$ is n .

Using this, prove that there are infinitely many primes p such that $p \equiv 1 \pmod{n}$.

Problem 4.54 Exam: Fall 2013, #4 | Difficulty: ★★ | Interest: ◇◇

Given a field K of characteristic p , when is an α that is algebraic over K said to be separable? Show that if α is algebraic over K , then α is separable if and only if $K(\alpha) = K(\alpha^{p^n})$ for all positive integers n .

Problem 4.55 Exam: Fall 2013, #8 | Difficulty: ★★ | Interest: ◇◇

Suppose that E/F is an algebraic extension of fields such that every nonconstant polynomial in $F[X]$ has at least one root in E . Show that E is algebraically closed.

Problem 4.56 Exam: Spring 2013, #3 | Difficulty: ★★ | Interest: ◇◇

Let $K \subset L$ be an algebraic extension of fields. An element a of L is called *abelian* if $K[a]$ is a Galois extension of K with abelian Galois group $\text{Gal}(K[a]/K)$. Show that the set of abelian elements of L is a subfield of L containing K .

Problem 4.57 Exam: Fall 2014, #3 | Difficulty: ★★ | Interest: ◇◇

Pick a nonzero rational number x . Determine all possibilities for the Galois group G of the normal closure of $\mathbb{Q}[\sqrt[4]{x}]$ over $\mathbb{Q}$, where $\sqrt[4]{x}$ is the root of $X^4 - x$ with maximal degree over $\mathbb{Q}$.

Problem 4.58 Exam: Spring 2015, #5 | Difficulty: ★★ | Interest: ◇◇

Let K and L be quadratic field extensions of a field k . Prove that $K \otimes_k L$ is an integral domain if and only if the k -algebras K and L are not isomorphic.

Problem 4.59 Exam: Spring 2015, #6 | Difficulty: ★★ | Interest: ◇◇

Let $K \subset L$ be subfields of $\mathbb{C}$ and let p be a prime. Assume K contains a non-trivial p -th root of unity. Show that L/K is a degree p Galois extension if and only if there is an element $a \in K$ that does not admit a p -th root, such that $L = K(\sqrt[p]{a})$.

Problem 4.60 Exam: Fall 2016, #5 | Difficulty: ★★ | Interest: ◇◇

Let $f \in F[X]$ be an irreducible separable polynomial of prime degree over a field F and let K/F be a splitting field of f . Prove that there is an element in the Galois group of K/F permuting cyclically all roots of f in K .

Problem 4.61 Exam: Fall 2016, #7 | Difficulty: ★★ | Interest: ◇◇

Let $f \in \mathbb{Q}[X]$ and $\xi \in \mathbb{C}$ a root of unity. Show that $f(\xi) \neq 2^{1/4}$.

Problem 4.62 Exam: Spring 2016, #6 | Difficulty: ★★ | Interest: ◇◇

Let K be a field of characteristic $p > 0$. For an element $a \in K$, show that the polynomial $P(X) = X^p - X + a$ is irreducible over K if and only if it has no root in K . Show also that if P is irreducible, then any root of it generates a cyclic extension of K of degree p .

Problem 4.63 Exam: Spring 2016, #10 | Difficulty: ★★ | Interest: ◇◇

(a) Determine the Galois group of the polynomial $X^4 - 2$ over $\mathbb{Q}$, as a subgroup of a permutation group. Also, give generators and relations for this group. (b) Determine the Galois group of the polynomial $X^3 - 3X - 1$ over $\mathbb{Q}$. (Hint: for polynomials of the form $X^3 + aX + b$, the discriminant $\Delta = -4a^3 - 27b^2$ plays a key role.) Explain your answer.

Problem 4.64 Exam: Fall 2017, #7 | Difficulty: ★★ | Interest: ◇◇

(a) Show that there is at most one extension $F(\alpha)$ of a field F such that $\alpha^4 \in F$, $\alpha^2 \notin F$, and $F(\alpha) = F(\alpha^2)$. (b) Find the isomorphism class of the Galois group of the splitting field of $x^4 - a$ for $a \in \mathbb{Q}$ with $a \notin \pm\mathbb{Q}^2$.

Problem 4.65 Exam: Spring 2017, #7 | Difficulty: ★★ | Interest: ◇◇

Find $[K : \mathbb{Q}]$ where K is a splitting field of $X^6 - 4X^3 + 1$ over $\mathbb{Q}$.

Problem 4.66 Exam: Fall 2018, #3 | Difficulty: ★★ | Interest: ◇◇

Let K/F be a finite extension of fields. Suppose that there exist finitely many intermediate fields $K \supset E \supset F$. Show that $K = F(x)$ for some $x \in K$.

Problem 4.67 Exam: Fall 2018, #12 | Difficulty: ★★ | Interest: ◇◇

Let F be a finite field and $K \subset \bar{F}$ the subfield of an algebraic closure generated by all roots of unity. Find all simple finite-dimensional K -algebras.

Problem 4.68 Exam: Spring 2018, #1 | Difficulty: ★★ | Interest: ◇◇

Let $\alpha \in \mathbb{C}$. Suppose that $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is finite and prime to $n!$ for an integer $n > 1$. Show that $\mathbb{Q}(\alpha^n) = \mathbb{Q}(\alpha)$.

Problem 4.69 Exam: Spring 2018, #8 | Difficulty: ★★ | Interest: ◇◇

Let F be a field that contains the real numbers $\mathbb{R}$ as a subfield. Show that the tensor product $F \otimes_{\mathbb{R}} \mathbb{C}$ is either a field or isomorphic to the product of two copies of F , $F \times F$.

Problem 4.70 Exam: Fall 2019, #2 | Difficulty: ★★ | Interest: ◇◇

Let L be a finite Galois extension of a field K inside an algebraic closure $\bar{K}$. Let M be a finite extension of K in $\bar{K}$. Show that the following are equivalent:

- (i) $L \cap M = K$,

- (ii) $[LM : K] = [L : K][M : K]$,
 (iii) every K -linearly independent subset of L is M -linearly independent.

Problem 4.71 Exam: Spring 2019, #5 | Difficulty: ★★ | Interest: ◇◇

Let Φ_n denote the n -th cyclotomic polynomial in $\mathbb{Z}[X]$ and let a be a positive integer and p a prime not dividing n . Prove that if $p \mid \Phi_n(a)$ in $\mathbb{Z}$, then $p \equiv 1 \pmod{n}$.

Problem 4.72 Exam: Fall 2020, #5 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and $f(X) = X^6 + 3 \in F[X]$. Determine a splitting field K of $f(X)$ over F and determine $[K : F]$ and $\text{Gal}(K/F)$ for each of the following three fields: $F = \mathbb{Q}$, $F = \mathbb{F}_5$, $F = \mathbb{F}_7$.

4.3 Challenging Problems (★★★)

Problem 4.73 Exam: Fall 2000, C3 | Difficulty: ★★★ | Interest: ◇◇◇

Let F be an infinite field of characteristic $p > 0$. Recall that a finite dimensional extension L/F is said to be simple if $L = F(u)$ for some element $u \in L$.

a) Suppose L/F has a finite number of intermediate fields. Show that L/F is simple. b) Let K be an intermediate field $F \subset K \subset L$, and suppose that $L = F(u)$ with $u^2 = a$, $a \in K$. Show that $K = F(a^{1/2}, \dots, a^{1/p})$. c) Conclude that L/F is simple if and only if there are a finite number of intermediate fields. d) Let $E = F(x, y)$ where x and y are indeterminates, and set $M = E(x^{1/p}, y^{1/p})$. Show that M/E has an infinite number of intermediate fields.

Problem 4.74 Exam: Fall 2001, F3 | Difficulty: ★★★ | Interest: ◇◇◇

Let G be the Galois group of $x^p - 2$ over $\mathbb{Q}$ where p is a prime. Show that G is isomorphic to the group of matrices of the form

$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$ where $a \in (\mathbb{Z}/p)^*$ and $b \in \mathbb{Z}/p$.

Problem 4.75 Exam: Winter 2002, C1 | Difficulty: ★★★ | Interest: ◇◇◇

The discriminant of the special cubic polynomial $f(x) = x^3 + ax + b$ is given by $-4a^3 - 27b^2$. Determine the Galois group of the splitting field of $x^3 - x + 1$ over (a) $\mathbb{F}_3$, the field with 3 elements. (b) $\mathbb{F}_5$, the field with 5 elements. (c) $\mathbb{Q}$, the rational numbers.

Problem 4.76 Exam: Fall 2003, PROBLEM 1 | Difficulty: ★★★ | Interest: ◇◇◇

Determine the Galois group of the polynomial $X^5 + 3X^2 + 1$ over $\mathbb{Q}$.

Problem 4.77 Exam: Winter 2005, D2 | Difficulty: ★★★ | Interest: ◇◇◇

Let F be a field of positive characteristic p and f the polynomial $x^p - x - a \in F[x]$. Let K/F be a splitting field of f . Show that K/F is Galois and determine explicitly (with proof) the Galois group of K/F .

Problem 4.78 Exam: Fall 2005, Sec.4 | Difficulty: ★★★ | Interest: ◇◇◇

- (a) Let F_q be the finite field with $q = p^n$ elements and let $N : F_q \rightarrow F_q$ be the norm map, defined by $N(z) = \prod_{\sigma \in \text{Gal}(F_q/F_p)} \sigma(z)$ where σ runs over the Galois group $G = \text{Gal}(F_q/F_p)$. Prove that N is surjective.
- (b) Let $v(x) = x^4 + ax^3 + bx^2 + cx + d$ be an irreducible polynomial of degree 4 in $\mathbb{Q}[x]$ and let K be the field generated by the complex roots $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ of v . Let F be the field generated by:
- $$B_1 = (\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4),$$
- $$B_2 = (\alpha_1 + \alpha_3)(\alpha_2 + \alpha_4),$$
- $$B_3 = (\alpha_1 + \alpha_4)(\alpha_2 + \alpha_3).$$
- Prove that K/F is an abelian extension, that is, the Galois group $H = \text{Gal}(K/F)$ is abelian. Hint: prove that the B_i are distinct and determine the possible elements of $\text{Gal}(K/\mathbb{Q})$ that fix them.
- (c) Let K be the splitting field of $p(x) = x^7 - 7$ over $\mathbb{Q}$. Describe the Galois group $G = \text{Gal}(K/\mathbb{Q})$ by giving generators and relations. Determine the number of quadratic subfields of K (a quadratic subfield is a subfield $E \subset K$ such that $[E : \mathbb{Q}] = 2$).

Problem 4.79 Exam: Spring 2006, F1 | Difficulty: ★★★ | Interest: ◇◇◇

- (a) Show that the Galois group of the splitting field of $X^8 - 2$ over $\mathbb{Q}$ has order 8. (b) Is this Galois group isomorphic to the dihedral group, the quaternion group or one of the three abelian groups of order 8?

Problem 4.80 Exam: Fall 2006, F2 | Difficulty: ★★★ | Interest: ◇◇◇

Let $f(x)$ be the polynomial $x^3 + 3$ over $\mathbb{Q}$ (the field of rational numbers). Determine the Galois group of $f(x)$, i.e., the Galois group of $K/\mathbb{Q}$ where K is a splitting field of $f(x)$.

Problem 4.81 Exam: Fall 2007, F2 | Difficulty: ★★★ | Interest: ◇◇◇

Let $f(x)$ be the polynomial $x^4 - 2x^2 - 2$ over $\mathbb{Q}$ and K be a splitting field of $f(x)$. Determine the Galois group $\text{Gal}(K/\mathbb{Q})$, and find the number of Galois extensions of $\mathbb{Q}$ inside K . Prove your answer.

Problem 4.82 Exam: Spring 2008, PROBLEM F1 | Difficulty: ★★★ | Interest: ◇◇◇

Consider the polynomial $P(X) = X^5 - 4X^2 + 2$ in $\mathbb{Q}[X]$. (a) Show that P is irreducible and has 3 real roots and 2 complex ones. (b) Show that the Galois group of P is S_5 .

Problem 4.83 Exam: Spring 2009, FI | Difficulty: ★★★ | Interest: ◇◇◇

Let $p(x) \in K[z]$ be a monic irreducible polynomial of degree n whose discriminant $D \neq 0$, where K has characteristic $\neq 2$. Prove that the Galois group of p is contained in the alternating group A_n if and only if $\sqrt{D} \in K$.

Problem 4.84 Exam: Spring 2010, Problem 12 | Difficulty: ★★★ | Interest: ◇◇◇

Determine all fields F such that the multiplicative group of F' is finitely generated.

Problem 4.85 Exam: Fall 2011, Problem 8 | Difficulty: ★★★ | Interest: ◇◇◇

Let I be the Galois group of the polynomial $X^5 - 9X + 3$ over $\mathbb{Q}$. Determine I . [Hint: Show that I contains an element of order 5 and that I contains a transposition, in a sense to be made precise.]

Problem 4.86 Exam: Spring 2013, #2 | Difficulty: ★★★ | Interest: ◇◇◇

Assume that L is a Galois extension of $\mathbb{Q}$ and that $K \subset L$ is the subfield generated by all roots of unity in L . Suppose that $L = \mathbb{Q}[a]$, where $a^n \in \mathbb{Q}$ for some positive integer n . Show that the Galois group $\text{Gal}(L/K)$ is cyclic.

Problem 4.87 Exam: Spring 2016, #7 | Difficulty: ★★★ | Interest: ◇◇◇

Show that for every positive integer n , there exists a cyclic extension of $\mathbb{Q}$ of degree n which is contained in $\mathbb{R}$.

Problem 4.88 Exam: Fall 2017, #8 | Difficulty: ★★★ | Interest: ◇◇◇

Let F be a field, and let $f, g \in F[x] \setminus \{0\}$ be relatively prime and not both constant. Show that $F(x)$ has finite degree $d = \max(\deg f, \deg g)$ over its subfield $F(f/g)$. (Hint: If the degree were less than d , show that there exists a nonzero polynomial with coefficients in $F[x]$ of degree less than d having f/g as a root.)

Problem 4.89 Exam: Fall 2018, #4 | Difficulty: ★★★ | Interest: ◇◇◇

Let K be a subfield of the real numbers and f an irreducible degree 4 polynomial over K . Suppose that f has exactly two real roots. Show that the Galois group of f is either S_4 or of order 8.

Problem 4.90 Exam: Spring 2018, #2 | Difficulty: ★★★ | Interest: ◇◇◇

Let $\zeta^9 = 1$ and $\zeta^3 \neq 1$ with $\zeta \in \mathbb{C}$.

(a) Show that $\sqrt[3]{3} \notin \mathbb{Q}(\zeta)$.

(b) If $\alpha^3 = 3$, show that α is not a cube in $\mathbb{Q}(\zeta, \alpha)$.

Problem 4.91 Exam: Fall 2019, #10 | Difficulty: ★★★ | Interest: ◇◇◇

Compute the Galois groups of the Galois closures of the following field extensions:

(a) $\mathbb{C}(x)/\mathbb{C}(x^4 + 1)$,

(b) $\mathbb{C}(x)/\mathbb{C}(x^4 + x^2 + 1)$,

where $\mathbb{C}(y)$ denotes the field of rational functions over $\mathbb{C}$ in a variable y .

Problem 4.92 Exam: Fall 2020, #6 | Difficulty: ★★★ | Interest: ◇◇◇

Let $K_1 \subset K_2 \subset K_3$ be fields with K_3/K_2 and K_2/K_1 both Galois. Let L be a minimal Galois extension of K_1 containing K_3 . Show that if the Galois groups $\text{Gal}(K_3/K_2)$ and $\text{Gal}(K_2/K_1)$ are both p -groups, then so is $\text{Gal}(L/K_1)$.

Problem 4.93 Exam: Spring 2020, #5 | Difficulty: ★★★ | Interest: ◇◇◇

If $K \neq \mathbb{Q}$ appears as a subfield (sharing the identity) of some central simple algebra over $\mathbb{Q}$ of $\mathbb{Q}$ -dimension 9, determine (isomorphism classes of) the groups appearing as the Galois group of the Galois closure of K over $\mathbb{Q}$.

Chapter 5

Linear Algebra

Problems on vector spaces, matrices, eigenvalues, Jordan and rational canonical forms, bilinear forms.

5.1 Standard Problems (★)

Problem 5.1 Exam: Fall 2000, D1 | Difficulty: ★ | Interest: ◇

Let V be a finite dimensional vector space over a field F and $T: V \rightarrow V$ a linear map whose characteristic polynomial has distinct roots, all in F . Show that T is diagonalizable.

Problem 5.2 Exam: Spring 2001, D1 | Difficulty: ★ | Interest: ◇

Let $V = M_{n \times n}(\mathbb{R})$ be the $n \times n$ real matrices. If $A, B \in V$, define $\langle A, B \rangle = \text{tr}(A'B)$. (a) Prove that $\langle, \rangle$ is a positive-definite symmetric inner product on V . (b) If E_{ij} is the matrix with a 1 in the i, j place and zeros elsewhere, prove that $\{E_{ij} \mid 1 \leq i, j \leq n\}$ is an orthonormal basis for V .

Problem 5.3 Exam: Fall 2001, LA1 | Difficulty: ★ | Interest: ◇

Let T be a linear operator on a finite-dimensional vector space V such that $\text{Im}(T)$ and $\text{Im}(T^m)$ have the same dimension. Show that $\ker T \cap \text{Im}(T) = 0$.

5.2 Moderate Problems (★★)

Problem 5.4 Exam: Spring 2000, D1 | Difficulty: ★★ | Interest: ◇◇

Let A be a linear operator on a vector space of dimension n such that A^m is the zero operator for some m .

- (a) Prove that all eigenvalues of A are equal to zero.
- (b) Prove that $A^m = 0$.

Problem 5.5 Exam: Spring 2000, D2 | Difficulty: ★★ | Interest: ◇◇

Describe, up to similarity, all 4×4 matrices A over $\mathbb{Q}$ such that $A^4 = -A^3$ but $A \neq 0$. Justify your answer.

Problem 5.6 Exam: Spring 2000, D3 | Difficulty: ★★ | Interest: ◇◇

Let $\text{End}(V)$ be the ring of all linear operators on a finite dimensional vector space V (with respect to the addition and composition of operators). For an operator $A \in \text{End}(V)$ let D_A be the subspace in $\text{End}(V)$ generated by the powers A^i , $i \geq 0$.

- Show that D_A is a subring in $\text{End}(V)$.
- Prove that if D_A is a field then the characteristic polynomial of A is a power of an irreducible polynomial.

Problem 5.7 Exam: Fall 2000, D2 | Difficulty: ★★ | Interest: ◇◇

Describe all non-diagonalizable 4×4 matrices over $\mathbb{Q}$ such that $A^3 + A^2 = 0$ up to similarity. Justify your answer.

Problem 5.8 Exam: Spring 2001, D2 | Difficulty: ★★ | Interest: ◇◇

Let k be a field. Prove that $\{x^i y^j \mid i, j \geq 0\}$ is a basis for the k -vector space $k[z] \otimes k[y]$. Use this to show that $k[z] \otimes k[y] = k[z, y]$ as vector spaces over k .

Problem 5.9 Exam: Fall 2001, LA2 | Difficulty: ★★ | Interest: ◇◇

Find all similarity classes of 4×4 matrices A over $\mathbb{Q}$ such that $A^4 = A$ and $A \neq I$ but $A^2 = A$ (I is the identity 4×4 matrix).

Problem 5.10 Exam: Spring 2002, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Let $\gamma : M_3(\mathbb{Q}) \rightarrow M_3(\mathbb{Q})$ be the map sending m to $\gamma(m) = m^2 + 3m + 3$. Show that $\gamma(m) = 0$ for all $m \in M_3(\mathbb{Q})$.

Problem 5.11 Exam: Spring 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let A be a real matrix with column vectors $A_1, A_2, \dots, A_n$. If the A_i are mutually orthogonal, then $n \leq \text{rank } A \leq n$.

This follows because $|\det(A)| = |\det(A_1 \dots A_n)| = \prod_{i=1}^n |A_i|$ and $A^T A$ is a diagonal matrix with diagonal entries $|A_1|^2, |A_2|^2, \dots, |A_n|^2$. Prove that a general matrix satisfies the inequality $|\det A| \leq \prod_{i=1}^n |A_i|$.

Hint: apply the Gram-Schmidt orthogonalization process to the columns.

Problem 5.12 Exam: Spring 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let $T \in M_3(\mathbb{C})$ and let A_T be the centralizer of T in $M_3(\mathbb{C})$. Show that $\dim(A_T) \geq 3$ and describe (up to similarity) the linear transformations T such that $\dim(A_T) = 3$.

e ° GALOIS THEORY

Problem 5.13 Exam: Spring 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Show that there exists a Galois extension of $\mathbb{Q}$ of degree p for each prime p . State precisely all results which are needed to justify your answer.

Problem 5.14 Exam: Spring 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let $a = \sqrt{-1} + 2$ where $i = \sqrt{-1}$. (a) Compute the minimal polynomial of a over $\mathbb{Q}$. (b) Let F be the splitting field and compute the degree of F over $\mathbb{Q}$; (c) Show that F contains 3 quadratic extensions of $\mathbb{Q}$; (d) Use this information to determine the Galois group.

Problem 5.15 Exam: Winter 2002, D1 | Difficulty: ★★ | Interest: ◇◇

Let $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ where I is the $n \times n$ identity matrix. Suppose that S is a $2n \times 2n$ symplectic matrix, meaning that S is real and satisfies $SJS^T = J$, where S^T is the transpose of S . (a) Show that S is symplectic. (b) Show that S is similar to S^T . (c) It is always true that $\det S = 1$. Prove this in case $n = 1$.

Problem 5.16 Exam: Winter 2002, D2 | Difficulty: ★★ | Interest: ◇◇

Suppose that A is a linear operator on the vector space C^n and that $v \in C^n$ satisfies $(A - aI)v = 0$ for some $a \in C$, so that uv is a generalized eigenvector of A with eigenvalue a . Suppose that $|a| < 1$. Show that $\|A^m u\| \rightarrow 0$ as $m \rightarrow \infty$, where $\|\cdot\|$ is the Euclidean norm on C^n .

Problem 5.17 Exam: Winter 2002, D3 | Difficulty: ★★ | Interest: ◇◇

Let the $n \times n$ matrix A be defined over the field F . Suppose that A has finite order: $A^m = I$ for some positive integer m . (a) If the characteristic of F is 0, show that A may be diagonalized over F . (b) Show that the conclusion of (a) is not true for an arbitrary r field F .

Problem 5.18 Exam: Fall 2002, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

Let T be a linear operator on a finite-dimensional vector space V such that $\text{im } T = \text{im } T^2$. Prove that $\ker T = \ker T^2$. (im and ker refer to the image and kernel of T)

Problem 5.19 Exam: Fall 2002, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Determine, up to similarity, all 3×3 matrices A over $\mathbb{Q}$ such that $A^2 + 2A + A^* = 0$ but $A + A^* \neq 0$.

Problem 5.20 Exam: Fall 2002, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let $T_1, T_2, \dots, T_m$ be linear operators on a vector space of dimension n . Assume that (i) $\dim(\text{im } T_i) = 1$ and (ii) $T_i^2 = 0$ and $T_i T_j = 0$ for every $i \neq j$. that $m < n$.

Problem 5.21 Exam: Winter 2003, D1 | Difficulty: ★★ | Interest: ◇◇

Let L be the subgroup of $Z^?$ with generators $(3, 2, 1)$ and $(2, 2, 6)$. Represent the quotient group $A = Z^?/L$ as a product of cyclic groups.

Problem 5.22 Exam: Winter 2003, D2 | Difficulty: ★★ | Interest: ◇◇

List suitably the Jordan canonical forms of all matrices A that satisfy (i) $\text{tr}(A) = 1$, (ii) $\text{tr}(A)^? = 2\det(A) + \text{tr}(A^?)$.

Problem 5.23 Exam: Winter 2003, D3 | Difficulty: ★★ | Interest: ◇◇

Let M be a matrix with complex entries. Deduce, using the structure theory of modules, that: (i) $M = S + N$, where S is semisimple (i.e. diagonalizable), N is nilpotent, and N and S commute. (ii) $N = P(M)$ where $P(X)$ is a polynomial with complex coefficients.

Problem 5.24 Exam: Fall 2003, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇◇

Let A be a linear operator in a finite dimensional vector space. Prove that if $A^? = A$ then $\text{Trace}(A) = \text{Rank}(A)$.

Problem 5.25 Exam: Fall 2003, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Let L/F be a field extension and let A and B be $n \times n$ matrices over F . Prove that if A and B are conjugate in $M_{n,n}(L)$, then A and B are conjugate in $M_{n,n}(F)$.

Problem 5.26 Exam: Fall 2003, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇

Let V be a finite dimensional vector space over a field F . Let B be a non-degenerate bilinear form on V . (For every nonzero $v \in V$ there is $u \in V$ such that $B(u, v) \neq 0$.)

- For every $v \in V$ define a linear form $l_v : V \rightarrow F$ by $l_v(u) = B(u, v)$. Prove that the map $V \rightarrow V^*$ given by $v \mapsto l_v$, is an isomorphism of vector spaces.
- Prove that for every linear operator a of V there is a linear operator a^* such that $B(a(u), v) = B(u, a^*(v))$ for all $u, v \in V$.
- Prove that $(ab)^* = b^*a^*$ for every two linear operators a and b .
- Suppose that B is either symmetric or skew-symmetric (that is $B(u, v) = B(v, u)$ or $B(u, v) = -B(v, u)$ respectively). Prove that $a^{**} = a$ for a linear operator a .

Problem 5.27 Exam: Spring 2004, PROBLEM 1 | Difficulty: ★★ | Interest: ◇◇

- For which $z \in \mathbb{C}$ is $\begin{pmatrix} 1 & 2 \\ z-1 & 1 \end{pmatrix}$ not similar over $\mathbb{C}$ to a diagonal matrix? Justify your answer.
- Let J_n be the $n \times n$ matrix each of whose entries is 1. Determine those $n \in \mathbb{Z}^*$ for which J_n is diagonalizable over $\mathbb{C}$ and give a diagonal matrix that is similar to J_n for such n .

Problem 5.28 Exam: Spring 2004, PROBLEM 2 | Difficulty: ★★ | Interest: ◇◇

Find an explicit formula for the determinant of a 3×3 complex matrix A as a polynomial in the traces $\text{Tr}(A^n)$ for $n = 1, 2, \dots$

Problem 5.29 Exam: Spring 2004, PROBLEM 3 | Difficulty: ★★ | Interest: ◇◇◇

Let V be a vector space over $\mathbb{C}$ of dimension $d > 0$. Suppose that A, B, C are linear operators on V such that

$AB - BA = C$. Suppose also that C commutes with both A and B . If V has no proper non-zero subspace that is left stable under all three operators, show that $d=1$.

Problem 5.30 Exam: Winter 2005, B1 | Difficulty: ★★ | Interest: ◇◇

An $n \times n$ real symmetric matrix P is positive definite if the inner product $P(x, y) = x^T P y$ is positive definite (that is, $P(x, x) > 0$ for all $0 \neq x \in \mathbb{R}^n$). Let S be an $n \times n$ invertible real symmetric matrix. Let $W \subseteq \mathbb{R}^n$ be a subspace such that the inner product $S(x, y) = x^T S y$ is positive definite on W but S is not positive definite on $W + \mathbb{R}a$ for any $xz \notin W$. (a) Show that $\mathbb{R}^n = W \oplus W^\perp$ for $W^\perp = \{x \in \mathbb{R}^n \mid S(x, W) = 0\}$. (b) For each $x \in \mathbb{R}^n$, writing $x = tw + gw$ for $gw \in W$ and $tw \in W^\perp$, define $P(x, y) = S(tw, yw) - S(tw^\perp, yw)$. Show that $PS^{-1} = SP^{-1}$ and P is positive definite.

(c) If P is symmetric positive definite and satisfies $PS^{-1} = SP^{-1}$, there exists a subspace W such that $P = S$ on W and $P = -S$ on $W^\perp$.

Problem 5.31 Exam: Winter 2005, B2 | Difficulty: ★★ | Interest: ◇◇

Let V be a two dimensional vector space over a field F . Let $T: V \rightarrow V$ be a linear transformation of finite order m . Prove the following facts:

- (a) If $F = \mathbb{Q}$, then $m < 6$.
- (b) For any given positive integer N , there exists a finite field F and a nondiagonalizable T of order greater than N .

Problem 5.32 Exam: Winter 2005, B3 | Difficulty: ★★ | Interest: ◇◇

Let V be a finite dimensional vector space over a field F and $T: V \rightarrow V$ be a linear transformation. Let $v \in V$ be a non-zero vector in V . Prove the following facts:

- (a) There exists a monic polynomial $P(X)$ in $F[X]$ such that $P(T)v = 0$.
- (b) Among monic polynomials $P(X) \in F[X]$ with $P(T)v = 0$, there exists a unique polynomial $P_0(X)$ of minimal degree.
- (c) If $P(T)v = 0$, then $P_0(X)$ is a factor of $P(X)$ in $F[X]$.

Problem 5.33 Exam: Fall 2005, Sec.3 | Difficulty: ★★ | Interest: ◇◇

(a) Prove that if three distinct real numbers α_i and three arbitrary numbers β_i are given, then there exists a unique polynomial $f(x) \in \mathbb{R}[x]$ of degree at most 2 such that $f(\alpha_i) = \beta_i$.

(b) Let K/F be a field extension of finite degree n and assume $K = F(a)$ where a satisfies a polynomial $f(x)$ of degree n in $F[x]$. Let $g: K \rightarrow K$ be the F -linear map $g(r) = ar$. Show that the eigenvalues of g coincide with the roots of $f(z)$.

(c) A bilinear form $A(z, y)$ on a vector space V over C is called alternating if $A(v, w) = -A(w, v)$ for all $v, w \in V$ and it is called non-degenerate if, for each nonzero $v \in V$ there exists $w \in V$ such that $A(v, w) \neq 0$.

- (i) Prove that any two non-degenerate alternating forms on $V = C^n$ differ by a scalar multiple.
 (ii) Does this remain true for $V = C^n$ for $n > 2$? Either prove true or give a counterexample.

© 1 @ 2

Problem 5.34 Exam: Spring 2007, Problem 1 | Difficulty: ★★ | Interest: ◇◇

Let A be a linear operator in a Q -vector space V of dimension n such that the minimal polynomial of A has degree n . Prove that every linear operator on V that commutes with A is a polynomial in A over Q .

Problem 5.35 Exam: Spring 2007, Problem 2 | Difficulty: ★★ | Interest: ◇◇

Let $G = GL_n(C)$ be the multiplicative group of invertible $n \times n$ matrices over C . Prove that every element of finite order in G is conjugate to a diagonal matrix.

Problem 5.36 Exam: Spring 2007, Problem 3 | Difficulty: ★★ | Interest: ◇◇

Let $A(z, y)$ be a bilinear form on a vector space V of finite dimension and $V_1 = \{x \in V \mid A(x, y) = 0 \text{ for all } y \in V\}$, $V_2 = \{y \in V \mid A(x, y) = 0 \text{ for all } x \in V\}$. Prove that $\dim V_1 = \dim V_2$.

Problem 5.37 Exam: Fall 2007, LA3 | Difficulty: ★★ | Interest: ◇◇

Let F be a field of characteristic 0 and let Q be an invertible n by n symmetric matrix with entries in F .

- (i) Show that there exists an invertible matrix A such that AQA^T is diagonal.
 (ii) Does the same remain true if Q is not invertible? Explain.

Problem 5.38 Exam: Spring 2009, LA1 | Difficulty: ★★ | Interest: ◇◇

A matrix N is said to be nilpotent of order k if $N^k = 0$, but $N^{k-1} \neq 0$. If N is nilpotent of order k , prove that $k = \min\{m \mid \ker(N^m) = \ker(N^{m+1})\}$.

Problem 5.39 Exam: Spring 2009, LA2 | Difficulty: ★★ | Interest: ◇◇

Consider the matrix $A = \begin{pmatrix} 0 & 2 & -2 \\ -2 & 0 & 1 \\ 2 & -1 & 0 \end{pmatrix}$. Prove that $(I - A)(I + A)$ is an orthogonal matrix. (B) Compute $\det A$ as a function of $\det A$.

Problem 5.40 Exam: Spring 2009, LA3 | Difficulty: ★★ | Interest: ◇◇

Let $\sigma \in S_n$ be a permutation of n elements. Let P_σ be the $n \times n$ matrix taking the standard basis vector e_i to $e_{\sigma(i)}$ for all i . Describe the eigenvalues over C of P_σ , in terms of the cyclic decomposition of σ .

Problem 5.41 Exam: Spring 2011, Problem 8 | Difficulty: ★★ | Interest: ◇◇

Let V be a finite dimensional vector space over a field F . Prove that every right ideal in $\text{End}_F(V)$ is of the form $\{f : \text{im}(f) \subseteq W\}$ for a unique subspace W of V .

Problem 5.42 Exam: Spring 2011, Problem 9 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of invertible linear operators on a finite dimensional vector space V over the field of complex numbers. Prove that if $(\dim V)^2 > |G|$, then there is a proper non-zero subspace $W \subseteq V$ such that $g(w) \in W$ for every $g \in G$ and every $w \in W$.

Problem 5.43 Exam: Fall 2013, #7 | Difficulty: ★★ | Interest: ◇◇

Give an example of a 10×10 matrix over $\mathbb{R}$ with minimal polynomial $(X+1)^2(X^4+1)$ which is not similar to a matrix with rational coefficients.

Problem 5.44 Exam: Fall 2014, #5 | Difficulty: ★★ | Interest: ◇◇◇

Let R be a commutative algebra over $\mathbb{Q}$ of finite dimension n . Let $\rho : R \rightarrow M_n(\mathbb{Q})$ be the regular representation, and define $\text{Tr} : R \rightarrow \mathbb{Q}$ by the matrix trace of ρ . If the pairing $(x, y) = \text{Tr}(xy)$ is non-degenerate on R , prove that R is semisimple.

Problem 5.45 Exam: Spring 2017, #1 | Difficulty: ★★ | Interest: ◇◇

Choose a representative for every conjugacy class in the group $GL(2, \mathbb{R})$. Justify your answer.

5.3 Challenging Problems (★★★)

Problem 5.46 Exam: Fall 2000, D3 | Difficulty: ★★★ | Interest: ◇◇◇

Let V be a complex vector space with positive definite inner product $(\cdot, \cdot)$, and $T: V \rightarrow V$ a linear map. Recall that the adjoint T^* of T is defined by: $(T(x), y) = (x, T^*(y))$ for all $x, y \in V$, and that T is called normal if T and T^* commute. Suppose that $T^2 = TT^*$. Let U be the kernel of T and W the orthogonal complement of U in V . (a) Show that W is T^* -invariant. (b) Show that U is T^* -invariant. (c) Conclude that W is T invariant. (d) Show that the restrictions of T and T^* to W commute. (e) Show that T is normal.

Problem 5.47 Exam: Spring 2001, D3 | Difficulty: ★★★ | Interest: ◇◇◇

Let $SO(3) = \{A \in M_{3 \times 3}(\mathbb{R}) \mid A^t = -A \text{ and } \det(A) = 1\}$. (a) Prove if $A \in SO(3)$ then $+1$ is an eigenvalue of A . (b) Prove that if W is the subspace of $\mathbb{R}^3$ orthogonal to a non-zero eigenvector of $A \in SO(3)$ with eigenvalue $+1$ then A takes W to W and acts as rotation by some angle θ on W .

Problem 5.48 Exam: Fall 2001, LA3 | Difficulty: ★★★ | Interest: ◇◇◇

Let V be a vector space over a field k . A bilinear form $f: V \times V \rightarrow k$ is called skew-symmetric if $f(u, v) = -f(v, u)$ for all $u, v \in V$ and is called alternating if $f(v, v) = 0$ for all $v \in V$. (a) Prove that

every alternating form is skew-symmetric. (b) Give an example of a skew-symmetric form which is not alternating. Hint: choose k of characteristic 2. (c) Show that all alternating forms on V form a vector space $\text{Alt}(V)$ and find $\dim \text{Alt}(V)$ if $\dim V = n$.

Problem 5.49 Exam: Spring 2002, PROBLEM 1 | Difficulty: ★★★ | Interest: ◇◇

Let F_7 be the field with 7 elements and let ZL be the splitting field of the polynomial $X^{17} - 1 = 0$ over F_7 . Determine the degree of L over F_7 , explaining carefully the principles underlying your computation.

Problem 5.50 Exam: Fall 2007, LA2 | Difficulty: ★★★ | Interest: ◇◇

Let A be a 3 by 3 matrix with complex entries. Suppose that A satisfies the relation $A^2 + A + I_3 = 0$, where I_3 denotes the 3 by 3 identity matrix.

(i) List the possible Jordan normal forms of A . (ii) Suppose A has entries in $\mathbb{R}$. List the possible Jordan normal forms of A .

Chapter 6

Commutative Algebra

Problems on prime and maximal ideals, localization, Noetherian rings, Dedekind domains, integral extensions.

6.1 Standard Problems (★)

Problem 6.1 Exam: Spring 2001, B1 | Difficulty: ★ | Interest: ◇

A commutative ring R with unit is said to be a local ring if it has a unique maximal ideal. Show that a commutative ring R with unit is a local ring if and only if for any two elements $u, v \in R$ satisfying $u+v = 1$ at least one of u, v is a unit of R .

Problem 6.2 Exam: Spring 2016, #4 | Difficulty: ★ | Interest: ◇

Let A be a commutative ring, S a multiplicatively closed subset of A , and $A \rightarrow A[S^{-1}]$ the localization.

- (a) Which elements of A map to zero in $A[S^{-1}]$?
- (b) Let $\mathfrak{p}$ be a prime ideal in A . Show that the ideal generated by the image of $\mathfrak{p}$ in $A[S^{-1}]$ is prime if and only if $\mathfrak{p} \cap S = \emptyset$.

Problem 6.3 Exam: Spring 2017, #5 | Difficulty: ★ | Interest: ◇

Let S be a multiplicatively closed subset of a commutative ring R . For a prime ideal I in R with $I \cap S = \emptyset$, show that the ideal $I \cdot R[S^{-1}]$ in the localized ring $R[S^{-1}]$ is prime. Also show that sending I to $I \cdot R[S^{-1}]$ gives a bijection between the prime ideals in R that do not meet S and the prime ideals in the localized ring $R[S^{-1}]$.

Problem 6.4 Exam: Spring 2020, #4 | Difficulty: ★ | Interest: ◇

Compute the dimension of the tensor products $\mathbb{Q}[\sqrt{2}] \otimes_{\mathbb{Z}} \mathbb{Q}[\sqrt{2}]$ over $\mathbb{Q}$ and $\mathbb{Q}[\sqrt{2}] \otimes_{\mathbb{Z}} \mathbb{R}$ over $\mathbb{R}$. Is $\mathbb{R} \otimes_{\mathbb{Z}} \mathbb{R}$ finite-dimensional over $\mathbb{R}$?

6.2 Moderate Problems (★★)

Problem 6.5 Exam: Fall 2009, (C1) | Difficulty: ★★ | Interest: ◇◇

Find a homomorphism $A \rightarrow B$ of commutative rings (sending the identity of A to the identity of B) and non-zero A -modules M, N such that the canonical map $B \otimes_A \text{Hom}_A(M, N) \rightarrow \text{Hom}_B(B \otimes_A M, B \otimes_A N)$ is the zero map, and justify your answer. Prove that the map is an isomorphism if M is a finitely generated projective A -module.

Problem 6.6 Exam: Fall 2009, (C2) | Difficulty: ★★ | Interest: ◇◇

Prove the following facts: (a) Any subring of $\mathbb{Q}$ sharing the identity with $\mathbb{Q}$ is a PID. (b) For a subring $A \subseteq \mathbb{Z}[\frac{1}{p}]$ sharing the identity with $\mathbb{Z}[\frac{1}{p}]$, if $A \neq \mathbb{Z}[\frac{1}{p}]$, A is not a PID.

2 Part 5: Non-commutative Rings.

Problem 6.7 Exam: Fall 2009, (R1) | Difficulty: ★★ | Interest: ◇◇

Prove that every two-sided ideal of the ring $M_2(\mathbb{Z})$ is principal, i.e., generated by one element.

Problem 6.8 Exam: Spring 2010, Problem 7 | Difficulty: ★★ | Interest: ◇◇

Show that the group of units of the ring $\mathbb{Z}/N\mathbb{Z}$ is cyclic iff N is either a power of an odd prime number, twice a power of an odd prime number, or 4.

Problem 6.9 Exam: Spring 2010, Problem 8 | Difficulty: ★★ | Interest: ◇◇

Let F be the field with 2 elements and let $R = F[X]$. List up to isomorphism all R -modules with 8 elements that are cyclic.

Problem 6.10 Exam: Spring 2010, Problem 9 | Difficulty: ★★ | Interest: ◇◇

Let A be a left Noetherian ring. Show that every left invertible element $a \in A$ is two-sided invertible.

Problem 6.11 Exam: Spring 2010, Problem 10 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and V a finite-dimensional F -vector space. Show that $R = \text{End}_F(V)$ has no non-trivial two-sided ideals.

Problem 6.12 Exam: Fall 2010, Problem 7 | Difficulty: ★★ | Interest: ◇◇

Let k be a field and let f be an irreducible element of the polynomial ring $k[x, y]$.

(i) Describe the localization $k[x, y]_P$ where $P = (f)$. Prove that it is a subring of the field of rational functions $k(x, y)$.

(ii) For any $r \in k(x, y)$, prove that $r \in k[x, y]_P$ for all but finitely many choices of f .

Problem 6.13 Exam: Fall 2010, Problem 8 | Difficulty: ★★ | Interest: ◇◇

Let $\&$ be an algebraically closed field, $R = \&[y_1, \dots, y_n]$ and $f: R \rightarrow R^*$ be given by $f(p) = (pf_1, \dots, pf_n)$ with all $f_j \in R$. Let M be the ideal generated by $x, -a_1, \dots, -a_n$ in R^n , where $a_j \in \&$ for all j . Consider for an integer $r, r > 1$, the map $f_r: R^{\otimes r} \rightarrow (R/M)^{\otimes r}$ induced by f . Prove that $\ker(f_r) = 0$ if and only if $f_j(a_1, \dots, a_n) = 0$ for all j .

Problem 6.14 Exam: Fall 2010, Problem 9 | Difficulty: ★★ | Interest: ◇◇

Let R be a ring with 1. Show that the only two-sided ideals of $M_n(R)$, the $n \times n$ matrices with entries in R , are of the form $M_n(I)$ for some two-sided ideal I in R .

Problem 6.15 Exam: Fall 2010, Problem 10 | Difficulty: ★★ | Interest: ◇◇◇

Let G be a group and $\mathbb{Z}G$ the integral group ring, i.e. the set of formal finite sums $\sum_i n_i g_i$ where $n_i \in \mathbb{Z}$ and $g_i \in G$. Multiplication is $f \cdot h = \sum_{i,j} n_i m_j (g_i g_j)$. Let I be the two-sided ideal generated by $\{g - 1 \mid g \in G\}$. Construct a natural map $F: I/I^2 \rightarrow G/[G, G]$ and prove that this map is an isomorphism of $\mathbb{Z}$ -modules. . Part 6: Fields

Problem 6.16 Exam: Fall 2010, Problem 11 | Difficulty: ★★ | Interest: ◇◇

Show that the extension of $\mathbb{Q}$ generated by $\sqrt[5]{1} + \sqrt[3]{2}$ is equal to $\mathbb{Q}(\sqrt[5]{1}, \sqrt[3]{2})$.

Problem 6.17 Exam: Fall 2010, Problem 12 | Difficulty: ★★ | Interest: ◇◇◇

Show that the multiplicative group of a finite field is cyclic and use this result to prove that the polynomial $X^n + 1$ is never irreducible over any finite field. | 3

Problem 6.18 Exam: Spring 2012, #8 | Difficulty: ★★ | Interest: ◇◇

Let R be a reduced (meaning, no nonzero nilpotent elements) commutative ring that has a unique proper prime ideal. Show that R is a field.

Problem 6.19 Exam: Fall 2013, #6 | Difficulty: ★★ | Interest: ◇◇

Let I be an ideal of a commutative ring R and $a \in R$. Suppose the ideals $I + Ra$ and $(I : a) := \{x \in R \mid ax \in I\}$ are finitely generated. Prove that I is also finitely generated.

Problem 6.20 Exam: Fall 2016, #9 | Difficulty: ★★ | Interest: ◇◇

Let F be a field and $a \in F$. Prove that the functor from the category of commutative F -algebras to **Sets** taking an algebra R to the set of invertible elements of the ring $R[X]/(X^2 - a)$ is representable.

Problem 6.21 Exam: Spring 2016, #2 | Difficulty: ★★ | Interest: ◇◇

Consider the functor F from commutative rings to abelian groups that takes a commutative ring R to the group R^* of invertible elements. Does F have a left adjoint? Does F have a

right adjoint? Justify your answers.

Problem 6.22 Exam: Fall 2017, #4 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative Noetherian ring and A a finitely generated R -algebra (not necessarily commutative). Let B be an R -subalgebra of the center $Z(A)$. Assume A is a finitely generated B -module. Show that B is a finitely generated R -algebra.

Problem 6.23 Exam: Fall 2017, #6 | Difficulty: ★★ | Interest: ◇◇

Let R be an integral domain, and let M be an R -module. Prove that M is R -torsion-free if and only if the localization $M_{\mathfrak{p}}$ is $R_{\mathfrak{p}}$ -torsion-free for all prime ideals $\mathfrak{p}$ of R .

Problem 6.24 Exam: Spring 2017, #4 | Difficulty: ★★ | Interest: ◇◇

Show that the ring $R = \mathbb{C}[x, y]/(y^2 - x^3 + 1)$ is a Dedekind domain. (Hint: compare R with the subring $\mathbb{C}[x]$.)

Problem 6.25 Exam: Fall 2018, #5 | Difficulty: ★★ | Interest: ◇◇

Let R be a commutative ring. Show the following:

- Let S be a non-empty saturated multiplicative set in R , i.e., if $a, b \in R$ then $ab \in S$ if and only if $a, b \in S$. Show that $R \setminus S$ is a union of prime ideals.
- If R is a domain, show that R is a UFD if and only if every nonzero prime ideal in R contains a nonzero principal prime ideal.

Problem 6.26 Exam: Fall 2018, #6 | Difficulty: ★★ | Interest: ◇◇

Let A be an integrally closed Noetherian domain with quotient field F and K/F a finite separable field extension.

- If $\{x_1, \dots, x_n\}$ is a basis for K as an F -vector space, show that there exists $\{y_1, \dots, y_n\}$ in K such that $\text{Tr}_{K/F}(x_i y_j) = \delta_{i,j}$ for all i, j .
- If B is the integral closure of A in K , show that B is a finitely generated A -module.

Problem 6.27 Exam: Fall 2018, #8 | Difficulty: ★★ | Interest: ◇◇

Give an example of a diagram of commutative rings whose colimit in the category of commutative rings is different from its colimit in the larger category of rings (and ring homomorphisms).

Problem 6.28 Exam: Spring 2018, #7 | Difficulty: ★★ | Interest: ◇◇

Let B be a commutative Noetherian ring, and let A be a Noetherian subring of B . Let I be the nilradical of B . If B/I is finitely generated as an A -module, show that B is finitely generated as an A -module.

Problem 6.29 Exam: Spring 2018, #10 | Difficulty: ★★ | Interest: ◇◇

By one definition, a Dedekind domain is a commutative Noetherian integral domain R , integrally closed in its fraction field, such that R is not a field and every nonzero prime ideal in R is maximal. Let R be a Dedekind domain, and let S be a multiplicatively closed subset of R . Show that the localization $R[S^{-1}]$ is either the zero ring, a field, or a Dedekind domain.

Problem 6.30 Exam: Fall 2019, #3 | Difficulty: ★★ | Interest: ◇◇

Let I be the ideal $(x^2 - y^2 + z^2, (xy + 1)^2 - z, z^3)$ of $R = \mathbb{C}[x, y, z]$. Find the maximal ideals of R/I , as well as all of the points on the variety $V(I) = \{(a, b, c) \in \mathbb{C}^3 \mid f(a, b, c) = 0 \text{ for all } f \in I\}$.

Problem 6.31 Exam: Fall 2020, #7 | Difficulty: ★★ | Interest: ◇◇

Let R be a Dedekind domain with quotient field K and I a nonzero ideal in R . Show both of the following:

- (a) Every ideal in R/I is a principal ideal.
- (b) If J is a fractional ideal of R (i.e., $0 \neq J \subset K$ is an R -module with $dJ \subset R$ for some $d \in R$), then there exists $0 \neq x \in K$ such that $I + xJ = R$.

Problem 6.32 Exam: Fall 2020, #8 | Difficulty: ★★ | Interest: ◇◇

Consider $R = \mathbb{C}[X, Y]/(X^2, XY)$. Determine the prime ideals P of R . Which of the localizations R_P are integral domains?

Problem 6.33 Exam: Spring 2020, #3 | Difficulty: ★★ | Interest: ◇◇

Prove that a Noetherian commutative ring A is a finite ring if the following two conditions are satisfied:

- (i) the nilradical of A vanishes,
- (ii) localization at every maximal ideal is a finite ring.

6.3 Challenging Problems (★★★)

Problem 6.34 Exam: Fall 2009, (R2) | Difficulty: ★★★ | Interest: ◇◇◇

Let B be a central simple algebra over k of dimension 4 (so, the center of B is k and has no nontrivial two-sided ideals except for (0) and B itself). Prove the following facts.

- (a) All left ideals of B have even dimension. (b) $B \cong M_2(k)$ if and only if B is not a division algebra, where $M(k)$ is the matrix algebra of 2×2 matrices with coefficients in k .

Problem 6.35 Exam: Fall 2011, Problem 3 | Difficulty: ★★★ | Interest: ◇◇◇

Let A be a local UFD of Krull dimension 2 (meaning that the maximal integer m for which there exist strict inclusions of prime ideals $\mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \subsetneq \dots \subsetneq \mathfrak{p}_m$ in A is exactly 2). Let $\pi \in A$ be neither zero nor a unit. Prove that $R[[\pi]]$ is a PID.

Problem 6.36 Exam: Fall 2014, #2 | Difficulty: ★★★ | Interest: ◇◇◇

Let $\mathbb{F}_p$ denote the finite field of p elements. Consider the covariant functor F from commutative $\mathbb{F}_p$ -algebras to abelian groups sending a ring R to its p -th roots of unity $F(R) = \{\zeta \in R \mid \zeta^p = 1\}$.

- (a) Give an example of a finite local ring R such that $F(R)$ has p^2 elements.
- (b) Let $\text{Aut}(F)$ be the set of natural transformations of F into itself inducing a group automorphism of $F(A)$ for all commutative rings A . Prove that F is representable and use the Yoneda Lemma to compute $|\text{Aut}(F)|$.

Problem 6.37 Exam: Fall 2019, #9 | Difficulty: ★★★ | Interest: ◇◇◇

Let A be a unique factorization domain which is a $\mathbb{Q}$ -algebra. Let K be the fraction field of A . Let L be a quadratic extension field of K . Show that the integral closure of A in L is a finitely generated free A -module.

Problem 6.38 Exam: Spring 2019, #9 | Difficulty: ★★★ | Interest: ◇◇◇

- (a) Find a domain R and two nonzero elements $a, b \in R$ such that R equals the intersection of the localizations $R[1/a]$ and $R[1/b]$ (in the quotient field of R) and $aR + bR \neq R$. (b) Let $\mathcal{C}$ be the category of commutative rings. Prove that the functor $\mathcal{C} \rightarrow \mathbf{Sets}$ taking a commutative ring R to the set of all pairs $(a, b) \in R^2$ such that $aR + bR = R$ is not representable.

Chapter 7

Representation Theory

Problems on group representations, characters, character tables, Maschke's theorem, Schur's lemma.

7.1 Standard Problems (★)

Problem 7.1 Exam: Fall 2016, #4 | Difficulty: ★ | Interest: ◇

Let D be a dihedral group of order $2p$ with normal cyclic subgroup C of order p for an odd prime p . Find the number of n -dimensional irreducible representations of D (up to isomorphism) over $\mathbb{C}$ for each n , and justify your answer.

7.2 Moderate Problems (★★)

Problem 7.2 Exam: Fall 2009, (Rep 1) | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group and $\rho: G \rightarrow \text{GL}(V)$ a complex representation. Prove that (V, ρ) splits as a direct sum of irreducible representations of G . [Note: It does not suffice to just quote a theorem. You have to actually prove the statement.]

Problem 7.3 Exam: Fall 2009, (Rep 2) | Difficulty: ★★ | Interest: ◇◇

Let G be a finite p -group and $\rho: G \rightarrow \text{GL}(V)$ a representation in a $\mathbb{F}_p$ -vector space.
(a) Show that V has a one-dimensional G -invariant subspace W . (b) Show by example that (V, ρ) need not split into a direct sum of irreducible representations.

Problem 7.4 Exam: Spring 2010, Problem 5 | Difficulty: ★★ | Interest: ◇◇

Give the total number and the dimensions of the irreducible complex representations of S_n . Prove your answer.

Problem 7.5 Exam: Spring 2010, Problem 6 | Difficulty: ★★ | Interest: ◇◇

State and prove Schur's Lemma which describes $\text{Hom}(V, W)$ for V and W finite dimensional irreducible complex representations of a finite

group.

Problem 7.6 Exam: Fall 2010, Problem 5 | Difficulty: ★★ | Interest: ◇◇

Prove that if a finite group G acts transitively on a set S having more than one element then there exists an element of G which fixes no element of S . (Hint: first prove a formula that counts the average number of fixpoints of an arbitrary (i.e. not necessarily transitive) action.)

Problem 7.7 Exam: Fall 2010, Problem 6 | Difficulty: ★★ | Interest: ◇◇

Let R be the 5 dimensional tautological representation of S_5 . Show that R is isomorphic to the direct sum of the trivial representation and an irreducible 4 dimensional representation. State clearly any principles or theorems that you use.

Problem 7.8 Exam: Spring 2011, Problem 7 | Difficulty: ★★ | Interest: ◇◇

Let $T: V \rightarrow V$ be a linear operator on a finite dimensional vector space. Prove that the characteristic polynomial P_T is irreducible if and only if T has no nontrivial invariant subspaces.

Problem 7.9 Exam: Spring 2012, #1 | Difficulty: ★★ | Interest: ◇◇

Let V be a finite-dimensional vector space over $\mathbb{Q}$ and $G \subset GL(V)$ a finite subgroup. Prove that the $\mathbb{Q}$ -subalgebra of $\text{End}(V)$ generated by G is semisimple.

Problem 7.10 Exam: Spring 2012, #2 | Difficulty: ★★ | Interest: ◇◇

Let V be the vector space of all $a \in \mathbb{R}^n$ such that $a_1 + a_2 + \cdots + a_n = 0$. The symmetric group S_n acts naturally on V . Prove that the S_n -module V is simple.

Problem 7.11 Exam: Spring 2013, #10 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group of order n and $\rho: G \rightarrow GL(V)$ a complex representation of G of dimension n . Show that ρ cannot be irreducible.

Problem 7.12 Exam: Spring 2016, #8 | Difficulty: ★★ | Interest: ◇◇

Determine the character table of S_4 , the symmetric group on 4 letters. Justify your answer.

Problem 7.13 Exam: Spring 2017, #2 | Difficulty: ★★ | Interest: ◇◇

Let G be the group with presentation $\langle x, y : x^4 = 1, y^5 = 1, xyx^{-1} = y^2 \rangle$, which has order 20. Find the character table of G .

Problem 7.14 Exam: Fall 2018, #11 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group, ω be a primitive 3rd root of unity in $\mathbb{C}$, and suppose that the complex character table of G contains the row $1, \omega, \omega^2, 1$. Determine the whole complex character table of G , the order of the group, and the order of its conjugacy classes.

Problem 7.15 Exam: Spring 2018, #6 | Difficulty: ★★ | Interest: ◇◇

Let G be a group with a normal subgroup $N = \langle y, z \rangle \cong (\mathbb{Z}/2)^2$. Suppose that G has a subgroup $Q = \langle x \rangle \cong \mathbb{Z}/3$ such that the composition $Q \subset G \rightarrow G/N$ is an isomorphism. Suppose also that $xyx^{-1} = z$ and $xzx^{-1} = yz$. Compute the character table of G .

Problem 7.16 Exam: Fall 2019, #7 | Difficulty: ★★ | Interest: ◇◇

Let G be the group of order 12 with presentation $G = \langle g, h \mid g^4 = 1, h^3 = 1, ghg^{-1} = h^2 \rangle$. Find the conjugacy classes of G and the values of the characters of the irreducible complex representations of G of dimension greater than 1 on representatives of these classes.

Problem 7.17 Exam: Spring 2019, #8 | Difficulty: ★★ | Interest: ◇◇

Prove that every finite group of order n is isomorphic to a subgroup of $GL_{n-1}(\mathbb{C})$.

Problem 7.18 Exam: Fall 2020, #9 | Difficulty: ★★ | Interest: ◇◇

Let G be a finite group, F a field, and V a finite-dimensional F -vector space with $G \rightarrow GL(V)$ a faithful irreducible representation. Show that the center $Z(G)$ of G is cyclic.

7.3 Challenging Problems (★★★)

Problem 7.19 Exam: Spring 2001, A3 | Difficulty: ★★★ | Interest: ◇◇

Let G be a finitely generated group. a. Show for each integer n there exist finitely many subgroups of index n . b. Suppose that there exists a subgroup of finite index in G . Prove that G contains a characteristic subgroup of finite index. B. Rings.

Problem 7.20 Exam: Spring 2001, C3 | Difficulty: ★★★ | Interest: ◇◇◇

Prove that the polynomial $z^* + 1$ is not irreducible over any field of positive characteristic. D. Linear Algebra.

Problem 7.21 Exam: Fall 2013, #5 | Difficulty: ★★★ | Interest: ◇◇◇

Let G be a finite group with the property that for any element $g \in G$ of order n , and any integer r prime to n , the elements g and g^r lie in the same conjugacy class. Show that the character of every representation of G takes values in $\mathbb{Q}$ (in fact even in $\mathbb{Z}$). (Hint: Use Galois theory.)

Problem 7.22 Exam: Spring 2013, #7 | Difficulty: ★★★ | Interest: ◇◇◇

Let $F = \mathbb{F}_2$ be the field with two elements. Show that there is a ring homomorphism $F[GL_2(F)] \rightarrow M_2(F)$ sending the element g in the group ring to the matrix $g \in M_2(F)$. Show this homomorphism is surjective. Let K be the kernel; since it is a left ideal, it is a left $GL_2(F)$ -module. Is this module indecomposable? Describe the simple modules in its composition series.

Chapter 8

Homological Algebra

Problems on functors, adjoint functors, projective/injective resolutions, Ext, Tor, abelian categories.

8.1 Standard Problems (★)

Problem 8.1 Exam: Fall 2016, #8 | Difficulty: ★ | Interest: ◇

Prove that if a functor $F : \mathcal{C} \rightarrow \mathbf{Sets}$ has a left adjoint functor, then F is representable.

8.2 Moderate Problems (★★)

Problem 8.2 Exam: Fall 2009, (Cat 1) | Difficulty: ★★ | Interest: ◇◇

Let $\mathbf{Top}$ be the category of topological spaces. Recall that a morphism f in some category is called a monomorphism if, for any two morphisms g , and g_2 that can be precomposed with f , $fg = f g_2$ implies $g = g_2$. Dually, f is called an epimorphism if, for any g ; and g_2 that can be post-composed with f , $gf = g_2 f$ implies $g = g_2$.

(a) Show that a continuous map $f : X \rightarrow Y$ is a monomorphism in $\mathbf{Top}$ if and only if f is one-to-one. (b) Now show by example that an epimorphism in $\mathbf{Top}$ need not be onto.

Problem 8.3 Exam: Fall 2009, (Cat 2) | Difficulty: ★★ | Interest: ◇◇

Let $F : \mathbf{Ab} \rightarrow \mathbf{Sets}$ be the forgetful functor from abelian groups to sets. Show that F does not have a right adjoint.

Problem 8.4 Exam: Spring 2010, Problem 1 | Difficulty: ★★ | Interest: ◇◇

Show that the functor from (unitary) rings to groups sending a ring A to its group of units A^* is co-representable by a ring A . In other words, show there exists a ring A and a natural isomorphism of functors $\text{Hom}_{\mathbf{rings}}(R, A) \xrightarrow{\sim} A^*$.

Problem 8.5 Exam: Spring 2010, Problem 2 | Difficulty: ★★ | Interest: ◇◇

(i) Define what it means for two categories to be equivalent. (ii) A groupoid G is a category such that all morphisms are isomorphisms. G is called connected if for any two objects z and y , $\text{Hom}_G(z, y)$ is non-empty. Show that any non-empty connected groupoid is equivalent to a group, that is, a groupoid with one object.

Problem 8.6 Exam: Fall 2010, Problem 1 | Difficulty: ★★ | Interest: ◇◇

Let Grp be the category of groups and Ab the category of abelian groups. If $F: \text{Ab} \rightarrow \text{Grp}$ is the inclusion of categories, then find a left adjoint to F and prove that it is a left adjoint.

Problem 8.7 Exam: Fall 2010, Problem 2 | Difficulty: ★★ | Interest: ◇◇

Let $F: C \rightarrow \text{Sets}$ be a covariant functor on a category C . Assume that F' is representable by an object $C_r \in \text{Obj}(C)$. Identify which of the following statements are necessarily correct, proving your answers in each case. (i) If $C \in \text{Obj}(C)$ and $F(C) \neq G$, then there is an element $f \in \text{More}(C_p, C)$; (ii) If G is a left adjoint of F' , then G is representable, (i) If $C, D \in \text{Obj}(C)$, and there is a map of sets $f: F(C) \rightarrow F(D)$, then there exists a $g \in \text{More}(C, D)$. (iv) If $C, D \in \text{Obj}(C)$, and there exists a $g \in \text{More}(C, D)$, then there exists a map of sets $f: F(C) \rightarrow F(D)$ (note we are not guaranteed in advance that (v) If $A \in \text{More}(D, C_r)$, then for any $C \in \text{Obj}(C)$, there is a map of sets $f: F(C) \rightarrow \text{More}(D, C)$).

Problem 8.8 Exam: Fall 2011, Problem 6 | Difficulty: ★★ | Interest: ◇◇

Let A be a commutative ring and let M be a finitely generated A -module. Let $f: M \rightarrow M$ be F -linear such that $f \otimes \text{id} : M \otimes_R [T] \rightarrow M \otimes_R [T]$ is surjective. Prove that f is an isomorphism.

Problem 8.9 Exam: Fall 2011, Problem 7 | Difficulty: ★★ | Interest: ◇◇◇

Let C be the category of semi-symplectic topological quantum paramonoids of Rice-Paddy type, satisfying the Mussolini-Rostropovich equations at infinity. Let X, Y be objects of C such that the functors $\text{More}(X, -)$ and $\text{More}(Y, -)$ are isomorphic, as covariant functors from C to sets. Show that X and Y are isomorphic in C .

Problem 8.10 Exam: Spring 2012, #9 | Difficulty: ★★ | Interest: ◇◇

Let R be a flat commutative $\mathbb{Z}$ -algebra, and Mod_R the category of R -modules. Suppose that I is an injective R -module. Show that the underlying abelian group of I is divisible.

Problem 8.11 Exam: Spring 2013, #8 | Difficulty: ★★ | Interest: ◇◇

Let $\mathcal{C}$ and $\mathcal{D}$ be additive categories, and let $\Phi: \mathcal{C} \rightarrow \mathcal{D}$ be a functor. Show: if Φ has a right adjoint, then Φ commutes with direct sums and for any two objects $x, y \in \mathcal{C}$, the map $\Phi_{x,y}: \text{Hom}_{\mathcal{C}}(x, y) \rightarrow \text{Hom}_{\mathcal{D}}(\Phi x, \Phi y)$ is a homomorphism.

Problem 8.12 Exam: Fall 2017, #10 | Difficulty: ★★ | Interest: ◇◇

Let $\mathcal{C}$ be a category with finite products, and let $\mathcal{C}^2$ be the category of pairs of objects of $\mathcal{C}$ together with morphisms $(A, A') \rightarrow (B, B')$ consisting of pairs of morphisms $(A \rightarrow B, A' \rightarrow B')$. Let $F : \mathcal{C}^2 \rightarrow \mathcal{C}$ be the direct product functor.

- (a) Find a left adjoint to F .
- (b) For $\mathcal{C}$ the category of abelian groups, determine whether or not F has a right adjoint.

Problem 8.13 Exam: Fall 2018, #7 | Difficulty: ★★ | Interest: ◇◇

Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor with a right adjoint G . Show that F is fully faithful if and only if the unit of the adjunction $\eta : \text{Id}_{\mathcal{C}} \rightarrow GF$ is an isomorphism.

Problem 8.14 Exam: Spring 2019, #10 | Difficulty: ★★ | Interest: ◇◇

Let $\mathcal{C}$ be an abelian category. Prove that the following are equivalent:

- (1) Every object of $\mathcal{C}$ is projective.
- (2) Every object of $\mathcal{C}$ is injective.

Problem 8.15 Exam: Fall 2020, #10 | Difficulty: ★★ | Interest: ◇◇

Let $\mathcal{C}$ and $\mathcal{D}$ be categories, and suppose that every pair of morphisms in $\mathcal{C}$ admits a coequalizer. Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be a functor that preserves coequalizers. Suppose also that if h is a morphism in $\mathcal{C}$ such that $F(h)$ is an isomorphism, then h is an isomorphism. Show that F is faithful.

Appendix A

Problems by Exam

The following table lists all problems in this compilation organized by exam date.

Spring 2000

- A1 — *Group Theory*, ★
- A2 — *Group Theory*, ★★
- A3 — *Group Theory*, ★★★
- B1 — *Ring Theory*, ★
- B2 — *Ring Theory*, ★★
- B3 — *Ring Theory*, ★★★
- C2 — *Field Theory & Galois Theory*, ★★
- C3 — *Field Theory & Galois Theory*, ★★
- C1 — *Field Theory & Galois Theory*, ★★
- D1 — *Linear Algebra*, ★★
- D2 — *Linear Algebra*, ★★
- D3 — *Linear Algebra*, ★★

Fall 2000

- A2 — *Group Theory*, ★★
- A3 — *Group Theory*, ★★★
- A1 — *Group Theory*, ★
- B2 — *Ring Theory*, ★★
- B3 — *Ring Theory*, ★★★

- B1 — *Ring Theory*, ★
- C2 — *Field Theory & Galois Theory*, ★★
- C3 — *Field Theory & Galois Theory*, ★★★
- C1 — *Field Theory & Galois Theory*, ★
- D1 — *Linear Algebra*, ★
- D2 — *Linear Algebra*, ★★
- D3 — *Linear Algebra*, ★★★

Spring 2001

- A2 — *Group Theory*, ★★
- A3 — *Representation Theory*, ★★★
- A1 — *Group Theory*, ★
- B1 — *Commutative Algebra*, ★
- B2 — *Module Theory*, ★★
- B3 — *Group Theory*, ★★★
- C1 — *Field Theory & Galois Theory*, ★
- C2 — *Field Theory & Galois Theory*, ★★
- C3 — *Representation Theory*, ★★★
- D1 — *Linear Algebra*, ★
- D2 — *Linear Algebra*, ★★
- D3 — *Linear Algebra*, ★★★

Fall 2001

- F1 — *Field Theory & Galois Theory*, ★
- F2 — *Field Theory & Galois Theory*, ★★
- F3 — *Field Theory & Galois Theory*, ★★★
- G1 — *Group Theory*, ★
- G2 — *Group Theory*, ★★
- G8 — *Group Theory*, ★★★
- LA1 — *Linear Algebra*, ★

- LA2 — *Linear Algebra*, ★★
- LA3 — *Linear Algebra*, ★★★
- R2 — *Ring Theory*, ★★
- R3 — *Ring Theory*, ★★

Spring 2002

- PROBLEM 1 — *Group Theory*, ★★
- PROBLEM 1 — *Ring Theory*, ★★
- PROBLEM 1 — *Linear Algebra*, ★★
- PROBLEM 1 — *Linear Algebra*, ★★★
- PROBLEM 2 — *Group Theory*, ★★
- PROBLEM 2 — *Ring Theory*, ★★
- PROBLEM 2 — *Linear Algebra*, ★★
- PROBLEM 2 — *Linear Algebra*, ★★
- PROBLEM 3 — *Ring Theory*, ★★
- PROBLEM 3 — *Linear Algebra*, ★★
- PROBLEM 3 — *Linear Algebra*, ★★

Winter 2002

- A2 — *Group Theory*, ★★
- A8 — *Group Theory*, ★★
- A1 — *Group Theory*, ★★
- B1 — *Ring Theory*, ★★
- B2 — *Ring Theory*, ★★
- B3 — *Ring Theory*, ★★
- C2 — *Field Theory & Galois Theory*, ★★
- C3 — *Field Theory & Galois Theory*, ★★
- C1 — *Field Theory & Galois Theory*, ★★★
- D1 — *Linear Algebra*, ★★
- D2 — *Linear Algebra*, ★★
- D3 — *Linear Algebra*, ★★

Fall 2002

- PROBLEM 1 — *Ring Theory*, ★★
- PROBLEM 1 — *Linear Algebra*, ★★
- PROBLEM 2 — *Group Theory*, ★★
- PROBLEM 2 — *Ring Theory*, ★★
- PROBLEM 2 — *Linear Algebra*, ★★
- PROBLEM 3 — *Group Theory*, ★★
- PROBLEM 3 — *Ring Theory*, ★★
- PROBLEM 3 — *Field Theory & Galois Theory*, ★★
- PROBLEM 3 — *Linear Algebra*, ★★

Winter 2003

- A2 — *Group Theory*, ★★
- A3 — *Group Theory*, ★★
- A1 — *Group Theory*, ★★
- B1 — *Ring Theory*, ★★
- B2 — *Ring Theory*, ★★
- B3 — *Ring Theory*, ★★
- C2 — *Field Theory & Galois Theory*, ★★
- C3 — *Field Theory & Galois Theory*, ★★
- C1 — *Field Theory & Galois Theory*, ★★
- D1 — *Linear Algebra*, ★★
- D2 — *Linear Algebra*, ★★
- D3 — *Linear Algebra*, ★★

Fall 2003

- PROBLEM 1 — *Group Theory*, ★★
- PROBLEM 1 — *Field Theory & Galois Theory*, ★★ ★
- PROBLEM 1 — *Linear Algebra*, ★★
- PROBLEM 2 — *Group Theory*, ★★

- PROBLEM 2 — *Ring Theory*, ★★
- PROBLEM 2 — *Field Theory & Galois Theory*, ★★
- PROBLEM 2 — *Linear Algebra*, ★★
- PROBLEM 3 — *Group Theory*, ★★
- PROBLEM 3 — *Field Theory & Galois Theory*, ★★
- PROBLEM 3 — *Linear Algebra*, ★★

Spring 2004

- PROBLEM 1 — *Group Theory*, ★★
- PROBLEM 1 — *Ring Theory*, ★★
- PROBLEM 1 — *Linear Algebra*, ★★
- PROBLEM 1 — *Field Theory & Galois Theory*, ★★
- PROBLEM 2 — *Ring Theory*, ★★
- PROBLEM 2 — *Linear Algebra*, ★★
- PROBLEM 2 — *Field Theory & Galois Theory*, ★★
- PROBLEM 3 — *Group Theory*, ★★
- PROBLEM 3 — *Ring Theory*, ★★
- PROBLEM 3 — *Linear Algebra*, ★★
- PROBLEM 3 — *Field Theory & Galois Theory*, ★★

Fall 2004

- Problem 1 — *Group Theory*, ★★★
- Problem 2 — *Ring Theory*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Field Theory & Galois Theory*, ★★

Winter 2005

- A2 — *Group Theory*, ★★
- A3 — *Group Theory*, ★★
- A1 — *Group Theory*, ★★
- B2 — *Linear Algebra*, ★★
- B3 — *Linear Algebra*, ★★
- B1 — *Linear Algebra*, ★★
- C1 — *Ring Theory*, ★★
- C2 — *Ring Theory*, ★★
- C3 — *Ring Theory*, ★★
- D1 — *Field Theory & Galois Theory*, ★★
- D2 — *Field Theory & Galois Theory*, ★★★
- D3 — *Field Theory & Galois Theory*, ★★

Fall 2005

- Sec.1 — *Group Theory*, ★★
- Sec.2 — *Ring Theory*, ★★
- Sec.3 — *Linear Algebra*, ★★
- Sec.4 — *Field Theory & Galois Theory*, ★★★

Spring 2006

- F2 — *Field Theory & Galois Theory*, ★★
- F3 — *Field Theory & Galois Theory*, ★★
- F1 — *Field Theory & Galois Theory*, ★★★
- G2 — *Group Theory*, ★★
- G3 — *Group Theory*, ★★
- G1 — *Group Theory*, ★★
- R2 — *Ring Theory*, ★★
- R3 — *Ring Theory*, ★★
- R1 — *Ring Theory*, ★★

Fall 2006

- F1 — *Field Theory & Galois Theory*, ★★
- F2 — *Field Theory & Galois Theory*, ★★★
- F3 — *Field Theory & Galois Theory*, ★★
- G1 — *Group Theory*, ★★
- G2 — *Group Theory*, ★★
- G3 — *Group Theory*, ★★
- R2 — *Ring Theory*, ★★
- R3 — *Ring Theory*, ★★

Spring 2007

- Problem 1 — *Group Theory*, ★★
- Problem 1 — *Ring Theory*, ★★
- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 1 — *Linear Algebra*, ★★
- Problem 2 — *Ring Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Linear Algebra*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 3 — *Ring Theory*, ★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 3 — *Linear Algebra*, ★★

Fall 2007

- F2 — *Field Theory & Galois Theory*, ★★★
- F3 — *Field Theory & Galois Theory*, ★★
- F1 — *Field Theory & Galois Theory*, ★★
- G1 — *Group Theory*, ★★
- G2 — *Group Theory*, ★★
- G3 — *Group Theory*, ★★

- LA2 — *Linear Algebra*, ★★★
- LA3 — *Linear Algebra*, ★★
- R2 — *Ring Theory*, ★★
- R3 — *Ring Theory*, ★★★
- R1 — *Ring Theory*, ★★

Spring 2008

- PROBLEM F1 — *Field Theory & Galois Theory*, ★★★
- PROBLEM F2 — *Field Theory & Galois Theory*, ★★
- PROBLEM F3 — *Field Theory & Galois Theory*, ★★
- PROBLEM G1 — *Group Theory*, ★★
- PROBLEM G2 — *Group Theory*, ★★
- PROBLEM G3 — *Group Theory*, ★★
- PROBLEM R1 — *Ring Theory*, ★★
- PROBLEM R2 — *Ring Theory*, ★★
- PROBLEM R3 — *Ring Theory*, ★★

Fall 2008

- F2 — *Field Theory & Galois Theory*, ★★
- F3 — *Field Theory & Galois Theory*, ★★
- F1 — *Field Theory & Galois Theory*, ★★
- G1 — *Group Theory*, ★★
- G2 — *Group Theory*, ★★
- G3 — *Group Theory*, ★★
- R2 — *Ring Theory*, ★★
- R3 — *Ring Theory*, ★★
- R1 — *Ring Theory*, ★★

Spring 2009

- F2 — *Field Theory & Galois Theory*, ★★
- F3 — *Field Theory & Galois Theory*, ★★
- F1 — *Field Theory & Galois Theory*, ★★★
- G1 — *Group Theory*, ★★
- G2 — *Group Theory*, ★★
- G3 — *Group Theory*, ★★
- LA1 — *Linear Algebra*, ★★
- LA2 — *Linear Algebra*, ★★
- LA3 — *Linear Algebra*, ★★
- R1 — *Ring Theory*, ★★
- R2 — *Ring Theory*, ★★
- R3 — *Ring Theory*, ★★

Fall 2009

- (C2) — *Commutative Algebra*, ★★
- (Cat 1) — *Homological Algebra*, ★★
- (Cat 2) — *Homological Algebra*, ★★
- (C1) — *Commutative Algebra*, ★★
- (F1) — *Field Theory & Galois Theory*, ★★
- (F2) — *Field Theory & Galois Theory*, ★★
- (Gr 1) — *Group Theory*, ★★
- (Gr 2) — *Group Theory*, ★★
- (R1) — *Commutative Algebra*, ★★
- (R2) — *Commutative Algebra*, ★★★
- (Rep 1) — *Representation Theory*, ★★
- (Rep 2) — *Representation Theory*, ★★

Spring 2010

- Problem 1 — *Homological Algebra*, ★★
- Problem 10 — *Commutative Algebra*, ★★
- Problem 11 — *Field Theory & Galois Theory*, ★★
- Problem 12 — *Field Theory & Galois Theory*, ★★★
- Problem 2 — *Homological Algebra*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Group Theory*, ★★
- Problem 5 — *Representation Theory*, ★★
- Problem 6 — *Representation Theory*, ★★
- Problem 7 — *Commutative Algebra*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Commutative Algebra*, ★★

Fall 2010

- Problem 1 — *Homological Algebra*, ★★
- Problem 10 — *Commutative Algebra*, ★★
- Problem 11 — *Commutative Algebra*, ★★
- Problem 12 — *Commutative Algebra*, ★★
- Problem 2 — *Homological Algebra*, ★★
- Problem 2 — *Group Theory*, ★★
- Problem 4 — *Group Theory*, ★★
- Problem 5 — *Representation Theory*, ★★
- Problem 6 — *Representation Theory*, ★★
- Problem 7 — *Commutative Algebra*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Commutative Algebra*, ★★

Spring 2011

- Problem 1 — *Group Theory*, ★★
- Problem 10 — *Group Theory*, ★★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 6 — *Field Theory & Galois Theory*, ★★
- Problem 7 — *Representation Theory*, ★★
- Problem 8 — *Linear Algebra*, ★★
- Problem 9 — *Linear Algebra*, ★★

Fall 2011

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 10 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Commutative Algebra*, ★★
- Problem 4 — *Ring Theory*, ★★
- Problem 5 — *Field Theory & Galois Theory*, ★★
- Problem 6 — *Homological Algebra*, ★★
- Problem 7 — *Homological Algebra*, ★★
- Problem 8 — *Field Theory & Galois Theory*, ★★
- Problem 9 — *Group Theory*, ★★

Spring 2012

- Problem 1 — *Representation Theory*, ★★
- Problem 2 — *Representation Theory*, ★★
- Problem 3 — *Module Theory*, ★★
- Problem 4 — *Ring Theory*, ★★
- Problem 5 — *Group Theory*, ★
- Problem 6 — *Field Theory & Galois Theory*, ★★
- Problem 7 — *Field Theory & Galois Theory*, ★★

- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Homological Algebra*, ★★
- Problem 10 — *Group Theory*, ★

Fall 2013

- Problem 1 — *Group Theory*, ★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 4 — *Field Theory & Galois Theory*, ★★
- Problem 5 — *Representation Theory*, ★★★
- Problem 6 — *Commutative Algebra*, ★★
- Problem 7 — *Linear Algebra*, ★★
- Problem 8 — *Field Theory & Galois Theory*, ★★
- Problem 9 — *Ring Theory*, ★
- Problem 10 — *Ring Theory*, ★★

Spring 2013

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 4 — *Module Theory*, ★★
- Problem 5 — *Ring Theory*, ★
- Problem 6 — *Ring Theory*, ★★
- Problem 7 — *Representation Theory*, ★★★
- Problem 8 — *Homological Algebra*, ★★
- Problem 9 — *Ring Theory*, ★★
- Problem 10 — *Representation Theory*, ★★

Fall 2014

- Problem 1 — *Group Theory*, ★★★
- Problem 2 — *Commutative Algebra*, ★★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 4 — *Ring Theory*, ★★★
- Problem 5 — *Linear Algebra*, ★★
- Problem 6 — *Group Theory*, ★★
- Problem 7 — *Group Theory*, ★★★
- Problem 8 — *Ring Theory*, ★
- Problem 9 — *Module Theory*, ★★
- Problem 10 — *Ring Theory*, ★★

Spring 2015

- Problem 1 — *Group Theory*, ★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Ring Theory*, ★★
- Problem 4 — *Module Theory*, ★★
- Problem 5 — *Field Theory & Galois Theory*, ★★
- Problem 6 — *Field Theory & Galois Theory*, ★★
- Problem 7 — *Ring Theory*, ★★
- Problem 8 — *Group Theory*, ★
- Problem 9 — *Ring Theory*, ★★
- Problem 10 — *Group Theory*, ★★

Fall 2016

- Problem 1 — *Group Theory*, ★
- Problem 2 — *Ring Theory*, ★★★
- Problem 3 — *Module Theory*, ★★
- Problem 4 — *Representation Theory*, ★
- Problem 5 — *Field Theory & Galois Theory*, ★★

- Problem 6 — *Field Theory & Galois Theory*, ★
- Problem 7 — *Field Theory & Galois Theory*, ★★
- Problem 8 — *Homological Algebra*, ★
- Problem 9 — *Commutative Algebra*, ★★
- Problem 10 — *Ring Theory*, ★★

Spring 2016

- Problem 1 — *Ring Theory*, ★
- Problem 2 — *Commutative Algebra*, ★★
- Problem 3 — *Ring Theory*, ★★
- Problem 4 — *Commutative Algebra*, ★
- Problem 5 — *Ring Theory*, ★★★
- Problem 6 — *Field Theory & Galois Theory*, ★★
- Problem 7 — *Field Theory & Galois Theory*, ★★★
- Problem 8 — *Representation Theory*, ★★
- Problem 9 — *Group Theory*, ★★
- Problem 10 — *Field Theory & Galois Theory*, ★★

Fall 2017

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Ring Theory*, ★★★
- Problem 4 — *Commutative Algebra*, ★★
- Problem 5 — *Module Theory*, ★★★
- Problem 6 — *Commutative Algebra*, ★★
- Problem 7 — *Field Theory & Galois Theory*, ★★
- Problem 8 — *Field Theory & Galois Theory*, ★★★
- Problem 9 — *Module Theory*, ★★★
- Problem 10 — *Homological Algebra*, ★★

Spring 2017

- Problem 1 — *Linear Algebra*, ★★
- Problem 2 — *Representation Theory*, ★★
- Problem 3 — *Group Theory*, ★★★
- Problem 4 — *Commutative Algebra*, ★★
- Problem 5 — *Commutative Algebra*, ★
- Problem 6 — *Ring Theory*, ★★
- Problem 7 — *Field Theory & Galois Theory*, ★★
- Problem 8 — *Module Theory*, ★★
- Problem 9 — *Module Theory*, ★★
- Problem 10 — *Field Theory & Galois Theory*, ★

Fall 2018

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 4 — *Field Theory & Galois Theory*, ★★★
- Problem 5 — *Commutative Algebra*, ★★
- Problem 6 — *Commutative Algebra*, ★★
- Problem 7 — *Homological Algebra*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Module Theory*, ★★
- Problem 10 — *Module Theory*, ★★★
- Problem 11 — *Representation Theory*, ★★
- Problem 12 — *Field Theory & Galois Theory*, ★★

Spring 2018

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★★
- Problem 3 — *Ring Theory*, ★★
- Problem 4 — *Ring Theory*, ★★★
- Problem 5 — *Module Theory*, ★★
- Problem 6 — *Representation Theory*, ★★
- Problem 7 — *Commutative Algebra*, ★★
- Problem 8 — *Field Theory & Galois Theory*, ★★
- Problem 9 — *Group Theory*, ★★
- Problem 10 — *Commutative Algebra*, ★★

Fall 2019

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★
- Problem 3 — *Commutative Algebra*, ★★
- Problem 4 — *Module Theory*, ★★
- Problem 5 — *Module Theory*, ★★
- Problem 6 — *Group Theory*, ★★
- Problem 7 — *Representation Theory*, ★★
- Problem 8 — *Module Theory*, ★★
- Problem 9 — *Commutative Algebra*, ★★★
- Problem 10 — *Field Theory & Galois Theory*, ★★★

Spring 2019

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Module Theory*, ★★
- Problem 3 — *Ring Theory*, ★★
- Problem 4 — *Module Theory*, ★
- Problem 5 — *Field Theory & Galois Theory*, ★★

- Problem 6 — *Field Theory & Galois Theory*, ★
- Problem 7 — *Ring Theory*, ★★
- Problem 8 — *Representation Theory*, ★★
- Problem 9 — *Commutative Algebra*, ★★★
- Problem 10 — *Homological Algebra*, ★★

Fall 2020

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Group Theory*, ★★
- Problem 3 — *Ring Theory*, ★★★
- Problem 4 — *Module Theory*, ★
- Problem 5 — *Field Theory & Galois Theory*, ★★
- Problem 6 — *Field Theory & Galois Theory*, ★★★
- Problem 7 — *Commutative Algebra*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Representation Theory*, ★★
- Problem 10 — *Homological Algebra*, ★★

Spring 2020

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Group Theory*, ★★★
- Problem 3 — *Commutative Algebra*, ★★
- Problem 4 — *Commutative Algebra*, ★
- Problem 5 — *Field Theory & Galois Theory*, ★★★
- Problem 6 — *Group Theory*, ★★
- Problem 7 — *Group Theory*, ★★
- Problem 8 — *Module Theory*, ★★★
- Problem 9 — *Module Theory*, ★★
- Problem 10 — *Module Theory*, ★★

Fall 2021

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Module Theory*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Representation Theory*, ★★
- Problem 5 — *Homological Algebra*, ★★
- Problem 6 — *Representation Theory*, ★★
- Problem 7 — *Commutative Algebra*, ★★
- Problem 8 — *Module Theory*, ★★
- Problem 9 — *Group Theory*, ★★
- Problem 10 — *Ring Theory*, ★★

Spring 2021

- Problem 1 — *Module Theory*, ★★ ★
- Problem 2 — *Ring Theory*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Field Theory & Galois Theory*, ★★
- Problem 5 — *Representation Theory*, ★★
- Problem 6 — *Homological Algebra*, ★★
- Problem 7 — *Representation Theory*, ★★ ★
- Problem 8 — *Ring Theory*, ★
- Problem 9 — *Commutative Algebra*, ★★
- Problem 10 — *Module Theory*, ★★ ★

Fall 2022

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Group Theory*, ★★
- Problem 5 — *Representation Theory*, ★

- Problem 6 — *Representation Theory*, ★★
- Problem 7 — *Commutative Algebra*, ★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Group Theory*, ★★
- Problem 10 — *Ring Theory*, ★★

Spring 2022

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Module Theory*, ★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Homological Algebra*, ★★
- Problem 5 — *Representation Theory*, ★★
- Problem 6 — *Group Theory*, ★★
- Problem 7 — *Field Theory & Galois Theory*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Ring Theory*, ★★
- Problem 10 — *Field Theory & Galois Theory*, ★★

Fall 2023

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Commutative Algebra*, ★★
- Problem 3 — *Commutative Algebra*, ★★
- Problem 4 — *Field Theory & Galois Theory*, ★★
- Problem 5 — *Field Theory & Galois Theory*, ★★
- Problem 6 — *Homological Algebra*, ★★
- Problem 7 — *Module Theory*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Representation Theory*, ★★
- Problem 10 — *Representation Theory*, ★

Spring 2023

- Problem 1 — *Homological Algebra*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 4 — *Ring Theory*, ★★
- Problem 5 — *Module Theory*, ★★
- Problem 6 — *Commutative Algebra*, ★★
- Problem 7 — *Commutative Algebra*, ★
- Problem 8 — *Group Theory*, ★★
- Problem 9 — *Representation Theory*, ★★
- Problem 10 — *Ring Theory*, ★★

Fall 2024

- Problem 1 — *Representation Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★
- Problem 3 — *Module Theory*, ★★
- Problem 4 — *Linear Algebra*, ★★
- Problem 5 — *Commutative Algebra*, ★★
- Problem 6 — *Group Theory*, ★
- Problem 7 — *Field Theory & Galois Theory*, ★★
- Problem 8 — *Module Theory*, ★★
- Problem 9 — *Ring Theory*, ★★
- Problem 10 — *Representation Theory*, ★

Spring 2024

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Commutative Algebra*, ★★
- Problem 3 — *Field Theory & Galois Theory*, ★★
- Problem 4 — *Homological Algebra*, ★★
- Problem 5 — *Module Theory*, ★

- Problem 6 — *Group Theory*, ★
- Problem 7 — *Group Theory*, ★★
- Problem 8 — *Representation Theory*, ★★
- Problem 9 — *Ring Theory*, ★★
- Problem 10 — *Homological Algebra*, ★

Fall 2025

- Problem 1 — *Commutative Algebra*, ★★
- Problem 2 — *Module Theory*, ★★★
- Problem 3 — *Group Theory*, ★★
- Problem 4 — *Representation Theory*, ★★
- Problem 5 — *Homological Algebra*, ★★
- Problem 6 — *Homological Algebra*, ★★
- Problem 7 — *Ring Theory*, ★
- Problem 8 — *Module Theory*, ★★
- Problem 9 — *Module Theory*, ★★★
- Problem 10 — *Field Theory & Galois Theory*, ★★

Spring 2025

- Problem 1 — *Field Theory & Galois Theory*, ★★
- Problem 2 — *Ring Theory*, ★★
- Problem 3 — *Representation Theory*, ★★
- Problem 4 — *Commutative Algebra*, ★★
- Problem 5 — *Representation Theory*, ★★
- Problem 6 — *Module Theory*, ★★
- Problem 7 — *Group Theory*, ★★
- Problem 8 — *Commutative Algebra*, ★★
- Problem 9 — *Linear Algebra*, ★★
- Problem 10 — *Ring Theory*, ★★

Spring 2026

- Problem 1 — *Group Theory*, ★★
- Problem 2 — *Field Theory & Galois Theory*, ★★
- Problem 3 — *Module Theory*, ★★★
- Problem 4 — *Group Theory*, ★★
- Problem 5 — *Homological Algebra*, ★★
- Problem 6 — *Representation Theory*, ★★
- Problem 7 — *Field Theory & Galois Theory*, ★★
- Problem 8 — *Module Theory*, ★★
- Problem 9 — *Representation Theory*, ★
- Problem 10 — *Commutative Algebra*, ★

Appendix B

Top Problems by Interest

The following problems received a three-diamond interest rating ($\diamond\diamond\diamond$), indicating particularly beautiful, surprising, or cross-cutting arguments.

Group Theory

- Spring 2015, # 8
- Fall 2016, # 1
- Fall 2003, PROBLEM 1
- Spring 2004, PROBLEM 3
- Fall 2005, Sec.1
- Spring 2010, Problem 4
- Fall 2010, Problem 4
- Spring 2011, Problem 1
- Spring 2011, Problem 2
- Fall 2011, Problem 2
- Fall 2018, # 1
- Fall 2019, # 1
- Spring 2019, # 1
- Fall 2021, # 9
- Spring 2023, # 8
- Fall 2000, A3
- Fall 2001, G8
- Fall 2004, Problem 1
- Fall 2014, # 1

Ring Theory

- Spring 2004, PROBLEM 3
- Fall 2000, B3
- Fall 2014, # 4
- Fall 2016, # 2
- Spring 2016, # 5
- Fall 2017, # 3
- Spring 2018, # 4
- Fall 2020, # 3

Module Theory

- Spring 2020, # 10
- Fall 2017, # 5
- Fall 2017, # 9
- Fall 2018, # 10
- Spring 2020, # 8
- Spring 2021, # 1
- Spring 2021, # 10
- Fall 2025, # 9
- Spring 2026, # 3

Field Theory & Galois Theory

- Spring 2004, PROBLEM 3
- Fall 2013, # 3
- Fall 2022, # 2
- Spring 2024, # 1
- Fall 2000, C3
- Fall 2001, F3
- Winter 2002, C1
- Fall 2003, PROBLEM 1

- Winter 2005, D2
- Fall 2005, Sec.4
- Spring 2006, F1
- Fall 2006, F2
- Fall 2007, F2
- Spring 2008, PROBLEM F1
- Spring 2009, F1
- Spring 2010, Problem 12
- Fall 2011, Problem 8
- Spring 2013, # 2
- Spring 2016, # 7
- Fall 2017, # 8
- Fall 2018, # 4
- Spring 2018, # 2
- Fall 2019, # 10
- Fall 2020, # 6
- Spring 2020, # 5
- Spring 2022, # 10
- Fall 2023, # 4

Linear Algebra

- Fall 2003, PROBLEM 1
- Spring 2004, PROBLEM 3
- Fall 2014, # 5
- Fall 2000, D3
- Spring 2001, D3
- Fall 2001, LA3

Commutative Algebra

- Fall 2010, Problem 10
- Fall 2010, Problem 12
- Fall 2009, (R2)
- Fall 2011, Problem 3
- Fall 2014, # 2
- Fall 2019, # 9
- Spring 2019, # 9

Representation Theory

- Spring 2022, # 5
- Spring 2001, C3
- Fall 2013, # 5
- Spring 2013, # 7
- Spring 2021, # 7

Homological Algebra

- Fall 2011, Problem 7

Index of Problems by Exam

Fall 2000	Problem 4, 40
A2, 10	Fall 2005
A3, 19	Sec.1, 13
A1, 9	Sec.2, 25
B2, 22	Sec.3, 55
B3, 30	Sec.4, 47
B1, 21	Fall 2006
C2, 38	F1, 41
C3, 46	F2, 47
C1, 37	F3, 41
D1, 51	G1, 13
D2, 52	G2, 13
D3, 57	G3, 13
Fall 2001	R2, 26
F1, 37	R3, 26
F2, 38	Fall 2007
F3, 46	F2, 47
G1, 9	F3, 41
G2, 10	F1, 41
G8, 19	G1, 14
LA1, 51	G2, 14
LA2, 52	G3, 14
LA3, 57	LA2, 58
R2, 22	LA3, 56
R3, 22	R2, 26
Fall 2002	R3, 30
PROBLEM 1, 23, 53	R1, 26
PROBLEM 2, 11, 23, 53	Fall 2008
PROBLEM 3, 11, 23, 39, 53	F2, 42
Fall 2003	F3, 42
PROBLEM 1, 12, 46, 54	F1, 42
PROBLEM 2, 12, 24, 39, 54	G1, 15
PROBLEM 3, 12, 39, 54	G2, 15
Fall 2004	G3, 15
Problem 1, 19	R2, 27
Problem 2, 24	R3, 27
Problem 3, 12	R1, 27

Fall 2009	Problem 1, 20
(C2), 60	Problem 10, 28
(Cat 1), 69	Problem 2, 64
(Cat 2), 69	Problem 3, 44
(C1), 60	Problem 4, 30
(F1), 42	Problem 5, 57
(F2), 42	Problem 6, 17
(Gr 1), 15	Problem 7, 20
(Gr 2), 16	Problem 8, 21
(R1), 60	Problem 9, 34
(R2), 63	Fall 2016
(Rep 1), 65	Problem 1, 10
(Rep 2), 65	Problem 10, 29
Fall 2010	Problem 2, 30
Problem 1, 70	Problem 3, 34
Problem 10, 61	Problem 4, 65
Problem 11, 61	Problem 5, 44
Problem 12, 61	Problem 6, 37
Problem 2, 16, 70	Problem 7, 44
Problem 4, 16	Problem 8, 69
Problem 5, 66	Problem 9, 61
Problem 6, 66	Fall 2017
Problem 7, 60	Problem 1, 17
Problem 8, 61	Problem 10, 71
Problem 9, 61	Problem 2, 18
Fall 2011	Problem 3, 30
Problem 1, 43	Problem 4, 62
Problem 10, 43	Problem 5, 36
Problem 2, 16	Problem 6, 62
Problem 3, 64	Problem 7, 45
Problem 4, 28	Problem 8, 48
Problem 5, 43	Problem 9, 36
Problem 6, 70	Fall 2018
Problem 7, 70	Problem 1, 18
Problem 8, 48	Problem 10, 36
Problem 9, 17	Problem 11, 66
Fall 2013	Problem 12, 45
Problem 1, 10	Problem 2, 18
Problem 10, 28	Problem 3, 45
Problem 2, 17	Problem 4, 48
Problem 3, 43	Problem 5, 62
Problem 4, 44	Problem 6, 62
Problem 5, 67	Problem 7, 71
Problem 6, 61	Problem 8, 62
Problem 7, 57	Problem 9, 34
Problem 8, 44	Fall 2019
Problem 9, 21	Problem 1, 18
Fall 2014	Problem 10, 49

- Problem 2, 45
- Problem 3, 63
- Problem 4, 35
- Problem 5, 35
- Problem 6, 18
- Problem 7, 67
- Problem 8, 35
- Problem 9, 64
- Fall 2020
 - Problem 1, 18
 - Problem 10, 71
 - Problem 2, 18
 - Problem 3, 31
 - Problem 4, 33
 - Problem 5, 46
 - Problem 6, 49
 - Problem 7, 63
 - Problem 8, 63
 - Problem 9, 67
- Spring 2000
 - A1, 9
 - A2, 10
 - A3, 19
 - B1, 21
 - B2, 22
 - B3, 30
 - C2, 38
 - C3, 38
 - C1, 38
 - D1, 51
 - D2, 52
 - D3, 52
- Spring 2001
 - A2, 10
 - A3, 67
 - A1, 9
 - B1, 59
 - B2, 33
 - B3, 19
 - C1, 37
 - C2, 38
 - C3, 67
 - D1, 51
 - D2, 52
 - D3, 57
- Spring 2002
 - PROBLEM 1, 10, 22, 52, 58
 - PROBLEM 2, 11, 22, 52, 53
 - PROBLEM 3, 23, 52, 53
- Spring 2004
 - PROBLEM 1, 12, 24, 39, 54
 - PROBLEM 2, 24, 40, 55
 - PROBLEM 3, 12, 24, 40, 55
- Spring 2006
 - F2, 40
 - F3, 40
 - F1, 47
 - G2, 13
 - G3, 13
 - G1, 13
 - R2, 25
 - R3, 25
 - R1, 25
- Spring 2007
 - Problem 1, 14, 26, 41, 56
 - Problem 2, 26, 41, 56
 - Problem 3, 14, 26, 41, 56
- Spring 2008
 - PROBLEM F1, 47
 - PROBLEM F2, 41
 - PROBLEM F3, 42
 - PROBLEM G1, 14
 - PROBLEM G2, 14
 - PROBLEM G3, 14
 - PROBLEM R1, 27
 - PROBLEM R2, 27
 - PROBLEM R3, 27
- Spring 2009
 - F2, 42
 - F3, 42
 - F1, 48
 - G1, 15
 - G2, 15
 - G3, 15
 - LA1, 56
 - LA2, 56
 - LA3, 56
 - R1, 27
 - R2, 28
 - R3, 28
- Spring 2010
 - Problem 1, 69
 - Problem 10, 60
 - Problem 11, 43
 - Problem 12, 48

- Problem 2, 70
- Problem 3, 16
- Problem 4, 16
- Problem 5, 65
- Problem 6, 65
- Problem 7, 60
- Problem 8, 60
- Problem 9, 60
- Spring 2011
 - Problem 1, 16
 - Problem 10, 16
 - Problem 2, 16
 - Problem 3, 43
 - Problem 6, 43
 - Problem 7, 66
 - Problem 8, 57
 - Problem 9, 57
- Spring 2012
 - Problem 1, 66
 - Problem 10, 9
 - Problem 2, 66
 - Problem 3, 33
 - Problem 4, 28
 - Problem 5, 9
 - Problem 6, 43
 - Problem 7, 43
 - Problem 8, 61
 - Problem 9, 70
- Spring 2013
 - Problem 1, 17
 - Problem 10, 66
 - Problem 2, 48
 - Problem 3, 44
 - Problem 4, 33
 - Problem 5, 21
 - Problem 6, 28
 - Problem 7, 67
 - Problem 8, 70
 - Problem 9, 28
- Spring 2015
 - Problem 1, 10
 - Problem 10, 17
 - Problem 2, 17
 - Problem 3, 28
 - Problem 4, 34
 - Problem 5, 44
 - Problem 6, 44
 - Problem 7, 29
 - Problem 8, 10
 - Problem 9, 29
- Spring 2016
 - Problem 1, 22
 - Problem 10, 45
 - Problem 2, 61
 - Problem 3, 29
 - Problem 4, 59
 - Problem 5, 30
 - Problem 6, 45
 - Problem 7, 48
 - Problem 8, 66
 - Problem 9, 17
- Spring 2017
 - Problem 1, 57
 - Problem 10, 37
 - Problem 2, 66
 - Problem 3, 20
 - Problem 4, 62
 - Problem 5, 59
 - Problem 6, 29
 - Problem 7, 45
 - Problem 8, 34
 - Problem 9, 34
- Spring 2018
 - Problem 1, 45
 - Problem 10, 63
 - Problem 2, 48
 - Problem 3, 29
 - Problem 4, 31
 - Problem 5, 34
 - Problem 6, 67
 - Problem 7, 62
 - Problem 8, 45
 - Problem 9, 18
- Spring 2019
 - Problem 1, 18
 - Problem 10, 71
 - Problem 2, 35
 - Problem 3, 29
 - Problem 4, 33
 - Problem 5, 46
 - Problem 6, 37
 - Problem 7, 29
 - Problem 8, 67
 - Problem 9, 64
- Spring 2020
 - Problem 1, 19

Problem 10, 35	A3, 11
Problem 2, 20	A1, 11
Problem 3, 63	B1, 23
Problem 4, 59	B2, 24
Problem 5, 49	B3, 24
Problem 6, 19	C2, 39
Problem 7, 19	C3, 39
Problem 8, 36	Cl, 39
Problem 9, 35	D1, 54
	D2, 54
Winter 2002	D3, 54
A2, 11	Winter 2005
A8, 11	A2, 12
A1, 11	A3, 13
B1, 23	A1, 12
B2, 23	B2, 55
B3, 23	B3, 55
C2, 38	B1, 55
C3, 39	C1, 25
Cl, 46	C2, 25
D1, 53	C3, 25
D2, 53	D1, 40
D3, 53	D2, 47
Winter 2003	D3, 40
A2, 11	